

CHEMISTRY GRADE 10: INTRODUCTION TO CHEMISTRY

CBE Phenomenon-Driven Lesson Sequence — Sub-Strand 1.1 (10 Lessons)

SUB-STRAND OVERVIEW	
Grade Level	10
Subject	Chemistry
Strand	Strand 1.0: Inorganic Chemistry
Sub-Strand	Sub-Strand 1.1: Introduction to Chemistry
Total Duration	10 lessons × 40 minutes = 400 minutes total
Sub-Strand Content	– Definition of chemistry – Branches of chemistry – Careers in chemistry – Chemistry in our daily lives (agriculture, pharmaceutical industry, medicine, manufacturing industry, entertainment industry, nuclear chemistry, sports industry, food industry, energy among others) – Drug, prescription, dosage and substance use – Learner's rights and responsibilities to a safe and healthy learning environment
Learning Outcomes	By the end of the sub-strand, the learner should be able to: - explain the meaning of Chemistry as a field of science, - explore the role of Chemistry in day-to-day life, - examine the effects of drug and substance use in day-to-day life, - advocate for a safe and healthy learning environment.
Core Competencies	– Communication and Collaboration: The learner explores teamwork with peers by contributing to group decision making through recognising and acknowledging the value of others' ideas during discussions. – Self-efficacy: The learner effectively communicates while presenting the findings on career opportunities related to Chemistry and how gender stereotypes influence career choices in plenary. – Digital literacy: The learner uses digital technology effectively to search for information on various career opportunities related to Chemistry. – Citizenship: The learner demonstrates critical and constructive dialogue while brainstorming with peers on the consumer rights to information on the drug prescription, substance use and knowledge on their side effects for awareness.
Core Values	– Respect: The learner shows acceptance while appreciating the opinion of each member as they discuss in groups the various branches of Chemistry.
Science & Engineering Practices	– Asking questions and defining problems: Learners generate questions about why substances such as salt, sand, sulphur, iron filings, antacid and paracetamol behave differently and add them to the Driving Question Board. – Planning and carrying out investigations: Learners group substances by properties, mix solids with liquids, separate the mixtures and heat solids to gather evidence.

	– Developing and using models: Learners build and revise a class model linking the kind of matter, its properties and its uses in daily life and medicine. – Analyzing and interpreting data: Learners interpret observation tables of substance properties and drug label information. – Constructing explanations: Learners explain why substances are used the way they are, such as why paracetamol is dosed and antacid is taken for acidity. – Obtaining, evaluating and communicating information: Learners search print and digital media on careers in chemistry and communicate findings through posters and plenary presentations.
Pertinent & Contemporary Issues (PCIs)	– Life skills (safety and security): Learners brainstorm their rights and responsibilities to a safe and healthy learning environment, including laboratory safety rules. – Health Promotion Issues (drug and substance use): Learners discuss the meaning of drug, prescription, dosage and substance use and their effects in day-to-day life. – Socio-Economic and Environmental issues (consumer protection): Learners explore labels on containers of manufactured products for marks of quality, expiry dates, nutritional information, preservatives and messages of caution.
Career Connections	– Pharmacist / Pharmaceutical technologist: develops, dispenses and advises on medicines such as antacids and paracetamol (e.g. Kenyan pharmaceutical firms in Nairobi). – Agricultural chemist / agronomist: fertilizers and soil testing for maize and tea farming in Kericho and the Rift Valley. – Food technologist: preservation, quality and labelling of food products such as milk, flour and juices. – Medical laboratory technologist and doctor: diagnosis and treatment of chronic diseases. – Chemical engineer / industrial chemist: manufacturing of detergents, soda ash at Lake Magadi, and cement. – Nuclear and energy chemist: nuclear energy and geothermal energy at Olkaria. – Sports scientist: anti-doping testing for athletes. – Forensic scientist: analysis of substances in drug-related investigations.
Focus for Lessons	This unit introduces Chemistry as the study of matter, its properties, composition and changes, and shows how it shapes medicine, food, manufacturing and daily life in Kenya. Learners start by grouping everyday substances and observing what happens when they are mixed, separated and heated, then build toward the branches, importance and careers of chemistry, and the responsible use of drugs. Instruction follows a predict-observe-explain cycle anchored on the Driving Question Board and a cumulative class model.
Driving Question / Key Inquiry Questions	DRIVING QUESTION: How do the different properties of substances around us help chemists make medicines, foods and other products, and how do we use them safely? KICD KEY INQUIRY QUESTIONS: 1. How are medicines for chronic diseases developed? 2. Do all substances around the world have the same properties and uses? 3. Why are food products, detergents and the things in the surrounding especially those that are man-made the way they are?
Anchoring Phenomenon	A tray at a school laboratory in Kisumu holds eight ordinary substances: table salt, water, sand, sulphur, iron filings, an antacid tablet and a paracetamol tablet. Learners are asked to sort them into groups with similar characteristics. They then mix solids with water: salt vanishes into the water, sand sinks and stays visible, iron filings are pulled out by a magnet, sulphur floats and refuses to wet, and the tablets fizz or crumble differently. Heating the solids gives different results: some melt, some burn with a blue flame and a sharp smell, and some change colour or char. Puzzling questions follow. Why do substances that all look like powders behave so differently? Why can some mixtures be separated easily while others cannot? How does knowing these properties help chemists make medicines for chronic diseases, food products and detergents with the properties they have? Learners must puzzle

	out why the antacid relieves heartburn while paracetamol relieves pain, and why taking too much of either is dangerous.
Storyline Thread	Lesson 1: Phenomenon – Learners predict, sort and observe salt, water, sand, sulphur, iron filings, antacid and paracetamol tablets, record their puzzling questions and start the Driving Question Board. Lesson 2: Matter – Learners mix solids with liquids, separate the mixtures and heat solids, then explain differences using properties and states of matter. Lesson 3: Matter (continued) – Learners classify substances by composition and properties, and use evidence to draw the first version of the class model. Lesson 4: Meaning of Chemistry as a field of science – Learners discuss with peers and define Chemistry, linking the definition to the evidence from the tray activity. Lesson 5: Branches of science and chemistry – In groups, learners explore organic, inorganic, physical, analytical and biochemistry and match each to the phenomenon, practising respect for others' opinions. Lesson 6: Importance of Chemistry – Learners brainstorm and explore chemistry in agriculture, pharmaceutical, medicine, manufacturing, entertainment, nuclear, food, sports and energy industries in Kenya, and revise the model. Lesson 7: Careers in chemistry – Learners search print and digital media for careers, discuss how gender stereotypes influence career choices and present findings in plenary. Lesson 8: Consumer protection and product labels – Learners examine labels on product containers for quality marks, expiry dates, preservatives and cautions. Lesson 9: Drugs and substance abuse – Learners discuss drug, prescription, dosage and substance use and examine their effects, then develop posters to sensitise peers and the community. Lesson 10: Safe and healthy learning environment and final model – Learners brainstorm their rights and responsibilities, present posters, review the Driving Question Board and finalise the model answering how properties of substances shape the products and medicines we use.
Supporting Phenomena	– Medicines for chronic diseases: how are drugs such as those for hypertension or diabetes, taken daily by many Kenyans, developed, and why must the dosage on a prescription be followed exactly? – Do all substances around the world have the same properties and uses? Compare salt from Lake Magadi and Mombasa salt pans, soda ash, sand and iron sheets used for roofing. – Why are food products, detergents and man-made things around us the way they are? Examine labels of unga, cooking oil, bar soap and juice packets showing expiry dates, preservatives and cautions. – Substance use: why do drugs such as alcohol, miraa and bhang affect the body and the learning environment differently from prescribed medicine?

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LESSON 1 (40 minutes (1 lesson)): The Kisumu Tray: Eight Substances, Many Puzzles

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	To launch the anchoring phenomenon. Learners observe and sort the eight substances on the Kisumu tray (A-H) by visible characteristics, record predictions about the water test, and write puzzling questions that start the class Driving Question Board (DQB).
Knowledge	Learners will be able to describe the visible appearance of substances A-H (table salt, water, sand, sulphur powder, iron filings, antacid tablet, paracetamol tablet, sugar) and explain that substances can be grouped in different ways using the characteristics we can see. Learners will recognise that appearance alone cannot tell us how a substance will behave. KICD outcome link (sub-strand 1.1): explore the role of Chemistry in day-to-day life.
Skills	Science and Engineering Practices: Asking Questions (writing testable and puzzling questions about the tray); Making Predictions and Developing Models (recording a prediction with a reason before any test); Planning and Carrying Out Investigations (careful, safe observation without tasting); Analysing Data (comparing sorting groups across the class). Core competency: Communication and Collaboration - contributing to group decisions by recognising and acknowledging the value of others' ideas.
Attitudes	Respect: the learner shows acceptance while appreciating the opinion of each group member when different sorting ideas are shared. Curiosity about everyday substances and care for safe laboratory practice.
Key Inquiry Question	KICD Key Inquiry Questions: (1) How are medicines for chronic diseases developed? (2) Do all substances around the world have the same properties and uses? (3) Why are food products, detergents and the things in the surrounding especially those that are man-made the way they are? Lesson 1 question: Substances A-H look alike in some ways, so why might they behave so differently?
Purpose in Storyline	This is the opening lesson and builds on no earlier lesson. It launches the phenomenon and the Driving Question: How do the different properties of substances around us help chemists make medicines, foods and other products, and how do we use them safely? Learners leave with their predictions, sorting groups and puzzling questions, which Lesson 2 will use as the starting point for testing mixing, separating and heating. The DQB started here will be revisited in later lessons. No explanations of why substances behave as they do are given in this lesson.
Safety Notes	No tasting or smelling directly from any substance, including the tablets and sugar, because laboratory items are never food or medicine. Goggles on for everyone during the tray handling. Wash hands after handling. Handle iron filings and sulphur powder with a spatula or in the labelled container and do not blow on powders. Tablets are for observation only and must not be swallowed. The water test is a prediction only today. Any later heating is teacher-led. Return all materials to the tray; wipe spills straight away.

B. LESSON OVERVIEW

Mrs Achieng walks into the Kisumu school laboratory with eight trays, one for each group. Each tray holds eight labelled substances: A table salt, B water (distilled, 50 mL in a beaker), C sand, D sulphur powder, E iron filings, F antacid tablet, G paracetamol tablet and H sugar. Lake Victoria is just down the road, and learners think about the water, sand and salt they know from home. Learners first predict what will happen when each solid is placed in water, then observe and sort the eight substances by what they can see, feel (through the container or spatula) and compare. They notice that powders that look alike might behave differently. Wanjiru asks why the antacid and paracetamol tablets might work differently, and Mrs Achieng records this on the class Driving Question Board. The lesson ends with each group writing at least three puzzling questions on sticky notes and posting them on the DQB under the Driving Question. No water test, heat test or mixture separation is carried out today; those are saved for Lesson 2. Learners observe, predict, wonder and begin to see why chemistry matters in daily life.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase  VIDEO: Types of mixtures <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  READING: <i>No closely matching reading in the ARES library for this activity.</i>  Search ARES for readings Search terms: chemistry property appearance substance	Learners sit in their groups of four (for example Group 1: Otieno, Wanjiru, Amina and Kipchirchir). Mrs Achieng shows the tray labelled A-H at the front but does not hand out materials yet. Learners individually write in their notebooks: (1) which of the solids A, C, D, E, F, G and H they think will dissolve in 50 mL of water at about 25 C, which will sink, which will float; (2) one reason for each prediction. They then share with their group and agree on one group prediction table with columns: Substance, Prediction (dissolves / sinks / floats / other), Reason. They are told that these are predictions only and that the actual test will be done in the next lesson. Learners also write one sentence on what they already know about how paracetamol and antacid tablets are used at home or in Kisumu clinics and pharmacies.	Hold up the tray and say: "This is the same kind of tray you will use today. Eight substances, eight letters. Before we touch anything, I want you to use what you already know from your kitchen, your shamba or the lakeshore." Display the question: "What will happen to each solid when I put one level spatula, about 2 g, into 50 mL of water at about 25 C?" Give 2 minutes of silent individual writing, then WAIT TIME of 10-15 seconds after asking: "What is one thing you are very sure about, and one thing you are not sure about?" before taking responses. Cold-call 4 learners from different groups (not volunteers only) to share one prediction and reason. Say: "Thank you, Kipchirchir. Who can add to or challenge that idea respectfully?" Do not say whether any prediction is right or wrong. Write 3-4 predictions on the board. Circulate and look at the reasons being written, noting any that mention floating, sinking, 'melting into the water' or 'vanishing'.	Predict-Observe-Explain (POE) - Predict step. Writing a prediction with a reason before any evidence makes learners' prior ideas visible to themselves and to the teacher. Later, when the class has evidence in Lesson 2, learners can compare it with these predictions and notice where their thinking changed. Individual writing before group talk gives every learner a voice (Respect).	Look for: (1) every learner has written a prediction and a reason for at least 5 of the 7 solids (A, C, D, E, F, G, H); (2) reasons are based on something real (for example 'salt goes into water when we make ugali water salty') rather than 'I just think so'; (3) the group table shows one agreed group prediction, with more than one learner's idea acknowledged. Note on a class list which learners offer no reason, and ask them a follow-up: "What have you seen at home that makes you think so?" Record common ideas on the board for use in Lesson 2.
Observe Phase  VIDEO: Solubility and intermolecular forces <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum)  Search ARES for similar videos	Wearing goggles, each group receives its tray with substances A-H in labelled containers. Learners take turns using a spatula and hand lens to observe and record each substance in a table: Substance letter, name, state (solid or liquid), colour, texture or grain size (fine, coarse, powder, crystals, tablet), shape, anything else they notice (for example shiny, dull, grains that stick together). Learners pass the containers around so each learner	Circulate to every group within the first 5 minutes. Ask: "What do you see? What do you notice that is the same? What is different?" Prompt Group 1: "Wanjiru, you picked up F and G. Tell me what looks alike and what looks different." Remind groups: "Goggles on, no tasting, no sniffing from the container. Use your eyes and the hand lens only." After 8 minutes ask groups to pause. Say: "Show me a different way to sort the same eight substances." WAIT	Sorting and Grouping (Patterns). Sorting by self-chosen rules helps learners see that the same set of substances can be grouped in many ways and that visible characteristics alone are not enough to predict behaviour. Making a second sort pushes learners to look for different features, preparing them to see the need for tests in Lesson 2. Gallery comparison across groups develops Communication and Collaboration.	Observable indicators: (1) every learner records at least 3 characteristics for each of substances A-H in the observation table; (2) each group produces two different sorting rules with the rule written next to each group; (3) learners use the correct names and letters (A table salt, B water, C sand, D sulphur powder, E iron filings, F antacid tablet, G paracetamol tablet, H sugar); (4) goggles are on and no tasting occurs. Ask a learner to explain

 READING: No closely matching reading in the ARES library for this activity.  Search ARES for readings Search terms: chemistry property appearance substance	observes all eight. They must not taste or smell directly. Learners then sort the eight substances into groups with similar characteristics using sorting cards labelled A-H. Each group chooses its own rule (for example 'white things', 'powders', 'tablets', 'things I use in food', 'things I do not think will dissolve') and records the groups. They then choose a second, different rule and sort again. Each group photographs or sketches both sorts on the group poster sheet. Learners notice puzzles, such as that salt, sugar and sand are all small grains yet may behave differently, and that antacid and paracetamol tablets both look white.	TIME of 10-15 seconds after the question "Can the same substance belong in two different groups? Think about it." Cold-call 3 groups to share one sorting rule (not their first), and ask the class: "Which substances ended up together in more than one group? Why?" Do not tell learners the 'right' way to group; there is none yet. Do not explain solubility, magnetism or heat behaviour. If a learner asks "Will it dissolve?", answer: "Good question. Put it on your question sticky note." Check that safety rules are being followed.		their group's rule, and check whether another learner could apply it to a new substance, such as a tablet or crystals. Note groups whose rules rely only on colour and push them: "What else could you use?"
Explain Phase  VIDEO: 01 Pure Substances Vs Mixtures <i>Related topic: not an exact match for this lesson.</i> Source: Seavuria  Search ARES for similar videos  READING: No closely matching reading in the ARES library for this activity.  Search ARES for readings Search terms: chemistry property appearance substance	Learners return to their predictions and sorting groups and make sense of what they have observed so far, without explaining the science of why. In groups they complete a 'What I notice / What puzzles me' T-chart. Under 'What I notice', they list evidence from the tray (for example 'A, C and H are all small white or pale grains', 'D is yellow powder', 'E is grey and gritty', 'F and G are both white tablets'). Under 'What puzzles me', they write questions, including at least one about how substances that look alike could behave differently, one about mixing, and one about the tablets. Wanjiru's question is shared: 'The antacid tablet and the paracetamol tablet both look like white tablets, so why does one help heartburn and the other help pain?' The class discusses why both are dangerous in excess, and why a label and the Pharmacy and Poisons Board matter, but without giving the	Ask: "Look at your sorting and your predictions. Is looking enough to know what a substance will do? Turn and talk to your partner for 1 minute." WAIT TIME of 10-15 seconds, then cold-call 4 learners from different groups. Probe: "What makes you say that?" and "What evidence from the tray supports that?" Read Wanjiru's question aloud and say: "Wanjiru has noticed something puzzling. Who has a similar puzzle? Raise your hand if you have wondered about medicines or about why salt and sand are so different." Take 3 more responses. Say clearly: "Today we are not explaining. We are noticing and wondering. In the next lesson we will test." Avoid giving any explanation of properties, dissolving, or how medicines work. Redirect: "Great thinking. Write that on a sticky note for our board." Link to Kenyan life: "Many of you have taken paracetamol or seen an antacid at	Noticing and Wondering (T-chart), with Think-Pair-Share. Separating observations from questions teaches learners to treat evidence and questions differently. Wanjiru's puzzle about the two tablets gives the class a shared, real-life question that can be traced from Lesson 1 to Lesson 10, linking the phenomenon to medicines, food and safety. Explaining is delayed on purpose so learners feel the need for evidence.	Observable indicators: (1) T-charts distinguish what was seen (observations) from what is wondered (questions); (2) at least 3 questions per group, with at least one linked to the tablets or to why powders behave differently; (3) learners say, in their own words, that appearance alone may not show how a substance behaves; (4) learners avoid giving explanations they have not tested. Ask 2 learners: "How is a question different from an observation?" Look for answers such as 'an observation is something I can see; a question is something I do not know yet.' Record who is ready to move to testing.

	answer. Learners also say where in Kenya they have seen these substances used: salt and sugar in ugali and chai (from Mumias or Nzoia sugar), sand in building around Kisumu, iron in tools, sulphur used on crops, tablets from a chemist. They briefly discuss: Is it enough to look at a substance to know what it will do?	a chemist in Kisumu. Why do you think the box has a label with a number of tablets and a warning?" Take 2 responses, and say that labels will return later. Record on chart paper: 'Looking is a start, but we may need tests.'		
Driving Question Board (DQB) Creation  VIDEO: Representing solutions using particulate models <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  READING: <i>No closely matching reading in the ARES library for this activity.</i>  Search ARES for readings Search terms: chemistry property appearance substance	Mrs Achieng posts the Driving Question on a large sheet at the front of the Kisumu lab: "How do the different properties of substances around us help chemists make medicines, foods and other products, and how do we use them safely?" Each group writes its best 3 puzzling questions on separate sticky notes, each with the group number. Learners come up one by one and post their notes on the DQB. As a class they cluster the questions into themes such as: (1) Why do powders behave differently? (2) Mixing and separating (for example 'Can salt be taken out of water?'). (3) Heating and burning (for example 'Why might sulphur burn?'). (4) Medicines, paracetamol, antacid (including Wanjiru's question). (5) Safety and labels ('Why is too much dangerous?'). Learners read the three KICD key inquiry questions and decide which themes link to them. A small 'Parking Lot' section is added for questions that do not fit.	Say: "This board belongs to us. Every question you add is evidence of your thinking. There are no silly questions." Model one example from Wanjiru: "Why does one white tablet help heartburn and another help pain?" Ask: "How are our questions connected to the Driving Question?" Give WAIT TIME of 10-15 seconds before taking answers. Cold-call 5 learners to read out and place a question on the board, ensuring all 8 groups are represented, including quieter learners such as Amina. Ask the class: "Which questions look alike? Where would we cluster them?" Invite one learner per cluster to suggest a heading. Ask: "Which of these questions can we investigate in the lab, and which need us to read or discuss?" Mark investigable ones with a star. Show the KICD key inquiry questions and ask: "Which of our questions fit here?" Tell learners that the board will be revisited every lesson and that the class model version 1 will be built in Lesson 3. Check that every learner has contributed at least one question and that the board is visible in the classroom.	Driving Question Board (DQB). The DQB turns individual curiosity into a shared, visible record of what the class wants to find out. Clustering questions into themes shows learners the structure of the topic and gives purpose to each later lesson. Returning to the board lets learners see their own progress from puzzled to explaining, which builds ownership and self-efficacy.	Observable indicators: (1) every group posts at least 3 questions, each written in full as a question; (2) at least half of the questions connect to the phenomenon (the tray A-H, tablets, mixing, heating, safety); (3) learners can justify a placement in a cluster with a reason; (4) learners listen respectfully when others explain a cluster (Respect). Ask 3 learners to point to a question from another group and say what they like about it. Record uncommon or advanced questions (for example about how chronic disease medicines are made) for later use. Take a photo of the DQB for the lesson record.
Model Building Phase	Learners begin an initial, personal 'Starting Ideas' sketch or concept map in their notebooks. At the	Say: "Draw what you think is going on. It is fine to be uncertain. Use question marks." Show a simple	Initial Modelling (Starting Ideas Sketch). Drawing ideas before any evidence creates a baseline model	Observable indicators: (1) each sketch has the centre 'Substances A-H', at least 2 ideas about why

 VIDEO: Types of mixtures <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  READING: <i>No closely matching reading in the ARES library for this activity.</i>  Search ARES for readings Search terms: chemistry property appearance substance	centre they write 'Substances A-H on the Kisumu tray'. They draw arrows to their current ideas about why the substances might behave differently, using only words and ideas they already have (for example 'what it is made of', 'size of grains', 'colour', 'heat'). They add a second arrow set from the tablets to 'helps the body' and 'danger if too much'. They mark with a question mark anything they are unsure of. This is not yet the class model; it is a record of where their thinking starts. Learners then share their starting ideas with a partner and add one new idea or question from the partner. Each learner keeps the sketch for revising in Lesson 3 when the first class model (v1) is built, after evidence is gathered in Lesson 2.	example frame on the board with the centre 'Substances A-H' and two empty arrows. Give 4 minutes of silent drawing, then WAIT TIME of 10-15 seconds before partner sharing. Circulate and ask: "What do you think makes salt different from sand? What makes you think so?" Cold-call 3 learners to share one idea from their sketch, including at least one with a question mark. Do not correct ideas. Say: "These are your starting ideas. In Lesson 2 you will gather evidence, and in Lesson 3 we will build our first class model, version 1, and compare it with this." Collect the sketches or ask learners to keep them safely in their notebooks. Before closing, ask learners to return all tray items, wipe benches and wash hands. Exit question: "Write one thing you are still wondering about the Kisumu tray."	that learners can revise later. Showing uncertainty with question marks normalises not knowing and gives the teacher information about the ideas that learners bring from home and from the lakeshore. Comparing this sketch with later models makes learning visible.	they differ, and at least one question mark; (2) the sketch links the tablets to both helping the body and danger in excess; (3) learners add at least one idea from a partner; (4) exit-ticket questions are specific (for example 'Why does sulphur float?'). Sort exit tickets into: ready to test, needs vocabulary support, and has advanced curiosity. Use these to plan groupings and prompts for Lesson 2.
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D. TEACHER REFLECTION

After the lesson, reflect: (1) Did every learner, including quieter ones such as Amina, contribute a prediction, an observation and a question? (2) Which sorting rules did groups invent, and which important features (such as grain size, shine or being a tablet) did they miss? (3) Did learners try to explain results too early, and how did I respond without giving the answer? (4) Which DQB clusters were strongest, and which learner questions should I use first in Lesson 2? (5) Were safety rules (goggles, no tasting, no sniffing, tablets not swallowed) followed throughout? (6) Did I give WAIT TIME of 10-15 seconds each time and cold-call the planned numbers (4, 3, 4, 5 and 3)? (7) What ideas about medicines did learners bring from home and community, and how will I handle them respectfully? Record the common predictions about substances A, C, D, E, F, G and H so that learners can compare them with real results in Lesson 2.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	What did we observe about substances A-H on the Kisumu tray? Record the colour, form (powder, crystals, grains, liquid, tablet) and any other visible features of A table salt, B water, C sand, D sulphur powder, E iron filings, F antacid tablet, G paracetamol tablet and H sugar, and the sorting rules our group used.
What did I learn?	What did we learn today? Write what we predicted for the water test and why, and what we realised about how looking alone may not tell us how a substance will behave. Also record the puzzling questions our group added to the Driving Question Board.
How does this explain the	How can we begin to explain the phenomenon? In your own words, write your starting idea about why substances that look alike

phenomenon?	(for example salt, sugar and sand, or the antacid and paracetamol tablets) might behave differently, and one thing you will need to test in Lesson 2 to check your idea.
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LESSON 2 (80 minutes (2 x 40-minute periods)): Matter: Mixing, Separating and Heating the Kisumu Tray

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	Learners gather evidence from the water test, the heat test and the separation of mixture M to explain that substances A-H behave differently because each has its own properties, which come from what it is made of. They use the particle idea of solids, liquids and gases to explain dissolving, separating and melting.
Knowledge	Matter is anything that has mass and occupies space. Solids have particles tightly packed with a fixed shape and volume. Liquids have particles close together but able to move, so they have a definite volume and take the shape of the container. Gases have particles far apart, moving independently and filling the whole container. A property is a characteristic used to identify or use a substance. Physical properties include appearance, solubility, magnetism, melting point and density. Chemical properties include burning and reacting with acid. Salt (A) and sugar (H) dissolve and cannot be filtered out. Sand (C), iron filings (E) and sulphur (D) do not dissolve. Iron is removed by magnetic separation, sand by filtration, and salt by evaporation.
Skills	NGSS Science and Engineering Practices: Planning and Carrying Out Investigations (water test, heat test and separating mixture M with safe procedures); Analyzing and Interpreting Data (comparing masses of 9 g in and 8.4 g out); Constructing Explanations (linking each result to a property and to particles); Developing and Using Models (particle drawings of solid, liquid and gas, and of dissolving); Engaging in Argument from Evidence. Core competency link: Communication and Collaboration, by contributing to group decisions and acknowledging the value of others' ideas.
Attitudes	Respect: learners accept and appreciate the opinion of each group member when they disagree about what the evidence shows. Learners show care and responsibility for safety in the laboratory, and curiosity about why everyday substances behave differently.
Key Inquiry Question	KICD Key Inquiry Question 2: Do all substances around the world have the same properties and uses? Also begins to answer Key Inquiry Question 3: Why are food products, detergents and the things in the surrounding the way they are?
Purpose in Storyline	Lesson 1 gave learners the sorting groups, water test predictions and puzzling questions about the Kisumu tray. This lesson supplies the first real evidence: it tests those predictions and shows that properties and states of matter explain the differences. It prepares Lesson 3, where learners will classify substances as elements, compounds or mixtures and build model v1 of the class DQB. It does not yet define chemistry (Lesson 4) or discuss the branches (Lesson 5).
Safety Notes	Goggles on for everyone throughout. No tasting or smelling directly: waft only when told to, and only the teacher notes the smell of burning sulphur, which gives off choking sulphur dioxide, so heat sulphur near an open window or in a fume cupboard. Heating is teacher-led only, with test tubes held in holders and pointed away from people. Tie back long hair, and keep the Bunsen flame away from learners' sleeves. Do not touch hot test tubes. Wash hands after handling iron filings, sulphur and tablets. The tablets A-H are for testing only and must never be swallowed. Wipe up spills of water on the floor immediately.

B. LESSON OVERVIEW

Mrs Achieng returns the Kisumu tray to each of the 8 groups. Learners revisit their Lesson 1 predictions and sorting groups, then test them: 1 level spatula (about 2 g) of each solid in 50 mL of water at about 25 C, followed by a teacher-led heat test and the separation of mixture M (3 g iron filings + 3 g sand + 3 g salt). They discover that substances that look alike behave very differently, then use properties and the particle ideas of solids, liquids and gases to explain what they saw, including why salt in water cannot be filtered out but sand can. They compare 9 g in and 8.4 g out to reason about loss in the procedure. Learners add new questions to the Driving Question Board (DQB) and sketch the first ideas that will feed model v1 in Lesson 3.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase  VIDEO: Video: States of Matter Source: PhET Interactive Simulations (English) Search ARES for similar videos  HTML: States of Matter: Basics Source: PhET Interactive Simulations (English) Search ARES for similar readings	Learners take out their Lesson 1 notebooks with the sorting groups and water test predictions for substances A-H (A table salt, B water, C sand, D sulphur powder, E iron filings, F antacid tablet, G paracetamol tablet, H sugar). Each learner privately writes a prediction for two new tests: (1) what happens to each solid in 50 mL of water, and (2) what happens when a small amount is heated in a dry test tube. They also predict whether mixture M (iron filings, sand and salt in water) can be separated and how many grams they expect to recover from 9 g. Then in Group 1 style fours (Otieno, Wanjiru, Amina and Kipchirchir are the model group), they compare predictions and record one agreement and one disagreement on a chart, giving a reason using 'because the substance is...'.	Say: 'Open your notebooks to your Lesson 1 sorting groups. Mrs Achieng's tray from Kisumu is back on your bench. Before we touch anything, I want your own prediction, in writing.' Ask: 'Which of the eight will disappear in water? Which will still be visible? Which will melt when heated? Write a reason, not just an answer.' Give WAIT TIME of 10-15 seconds of silent thinking before any talking. Circulate and cold-call 4 learners from different groups: 'Kipchirchir, what did you predict for E, the iron filings, and why?' Do not say whether predictions are right. Say: 'Wrong predictions are welcome, because they show us what to investigate.' Remind learners that no tasting is allowed. Point to a group that disagrees and say: 'Respect each other's ideas; we will let the evidence decide.' Record 3 contested predictions on the board, for example 'sulphur dissolves' and 'sugar melts like ice'.	Prediction with reasoning (commit-before-evidence): writing a prediction and a reason before testing makes learners' existing ideas visible and gives them something to confront with evidence. It builds understanding because learners must revise a specific claim they own, not simply receive an answer. It also links to Lesson 1 learning by reusing the sorting groups as a starting hypothesis about which substances behave alike.	Look for: (1) each learner has a written prediction for both tests with a reason; (2) at least 6 of 8 groups record one disagreement; (3) learners use words such as 'dissolve', 'melt', 'float' or 'sink' for at least 5 of the 8 substances; (4) the 4 cold-called learners can give a reason beyond 'I think so'. Note common ideas to address later: 'powders that look the same behave the same' and 'if it vanishes in water it has been destroyed'. Quick check: ask for a thumbs-up, sideways or down on 'I am sure about my prediction for the tablets' and note the groups showing thumbs down.
Observe Phase  VIDEO: Video: States of Matter: Basics Source: PhET Interactive Simulations (English) Search ARES for similar videos  HTML: States of Matter	Part A, water test (group work, about 20 minutes): for each solid A, C, D, E, F, G and H, learners add 1 level spatula (about 2 g) to 50 mL of water (B, distilled) at about 25 °C in a beaker, stir gently and record what they see. Expected observations: A salt dissolves completely to a clear colourless solution; C sand sinks, stays visible, water stays clear; D sulphur floats on top, is not wetted and stays yellow; E iron filings sink	Before starting say: 'Goggles on. Nobody tastes or smells anything directly. I do all the heating.' Model one test step by step for substance A, then say: 'Now you do C. Describe only what you can see, no explaining yet.' Move between the 8 groups and ask each one: 'What exactly did you see for D, the sulphur? Where is it in the beaker?' After the water test, give WAIT TIME of 10-15 seconds and cold-call 3 groups for results on	Structured observation table (data collection before explanation): learners fill a table of A-H with the same columns, forcing them to compare substances on the same tests. The table makes patterns visible, such as 'dissolves / does not dissolve' and 'melts / does not melt', before any explanation, which prevents learners from explaining things they did not see. The mass table for mixture M builds quantitative reasoning:	Observable indicators: (1) every group's table has entries for all 8 substances, including D burning with a blue flame and E glowing orange-red then dull blackish; (2) learners use 'dissolves' for A and H and 'does not dissolve' for C, D and E, not 'melts' or 'disappears'; (3) mass tables show $2.9 + 2.8 + 2.7 = 8.4$ g and a loss of 0.6 g; (4) learners can say that filtering salt solution does not remove salt; (5) safe behaviour: goggles on, no

Source: PhET Interactive Simulations (English) Search ARES for similar readings	and are pulled out by a magnet, not dissolved; F antacid tablet (used whole) crumbles into a cloudy white suspension and fizzes when dropped into dilute vinegar or lemon juice; G paracetamol tablet stays mostly intact, crumbles slowly, little dissolves, and does not fizz in vinegar; H sugar dissolves completely to a clear solution. Part B, mixture M separation: learners use a magnet to remove iron (recovered 2.9 g), filter the rest to remove sand (2.8 g after drying), then evaporate the filtrate to recover salt (2.7 g). They record the masses in a table, add them to get $2.9 + 2.8 + 2.7 = 8.4$ g and calculate the loss: $9 - 8.4 = 0.6$ g. They also try to filter a salt solution and find the salt passes through. Part C, heat test (teacher demonstration, learners observe from a safe distance with goggles on): a small amount of each solid in a dry test tube over a Bunsen flame. A salt crackles, no melting, no colour change; C sand no change; D sulphur melts to a runny yellow-brown liquid at about 115 C then burns with a blue flame and a sharp choking smell (sulphur dioxide); E iron filings glow orange-red and turn dull blackish; F antacid stays white and crumbles, no melting; G paracetamol melts at about 169 C and browns; H sugar melts at about 186 C, turns brown, then chars black with a caramel smell.	sand, salt and iron. Ask: 'Salt vanished and sand did not. Is the salt gone?' Prompt a quick check: have a learner filter salt solution and ask 'What came through the filter paper?' For the heat demonstration, stand at the front with the window open for sulphur and say: 'Watch the colour, the flame and whether it melts. Notice the smell only if I tell you; sulphur dioxide is choking.' Cold-call 3 more learners: 'Wanjiru, what happened to H, the sugar, as it got hotter?' Ensure each group has one data table of A-H, with columns for water test, magnet, heating and mass. Check the sums: 'Add 2.9, 2.8 and 2.7. What do you get? How much did we start with? Where could 0.6 g have gone?' Accept spillage and sticking to equipment as reasons, and do not accept 'destroyed'. Use Kenyan hooks: 'Think of how salt vanishes in your ugali pot and sugar in your chai.'	learners compare 9 g in with 8.4 g out and see that separation is real but not perfect.	tasting, test tubes pointed away. Use a checklist and note the groups that need a re-run, such as those that did not evaporate the filtrate to dryness. If a group records 'sulphur dissolved', send them back to look at where the sulphur is in the beaker.
Explain Phase  VIDEO: States of matter Source: Khan Academy (English - US curriculum)	Learners explain their evidence in three steps. First, in groups, they group A-H again using properties from the evidence (for example 'dissolves in water': A and H; 'attracted by a magnet': E; 'melts	Say: 'Put your Lesson 1 sorting next to your evidence table. Which groups still make sense and which need to change? Talk in your group for 3 minutes.' Ask: 'What does the evidence tell us about	Claim-Evidence-Reasoning (CER) with particle modelling: learners write a claim, back it with data from their own tables, then use the particle model of solids, liquids and gases as the reasoning that links	Look for: (1) CER paragraphs that cite at least two data points (for example 'salt dissolved to a clear solution' and 'sand stayed visible and was filtered, 2.8 g'); (2) correct use of 'dissolves' versus 'does not

Search ARES for similar videos HTML: Solid, Liquid and Gas: Phases of Matter (PLIX) Source: CK-12 Search ARES for similar readings	on heating': D, G and H; 'burns': D; 'fizzes with acid': F) and compare this with their Lesson 1 sorting groups, noting what changed. Second, they draw a particle picture of a solid (particles tightly packed with a fixed shape and volume), a liquid (particles close but moving around, definite volume, takes the shape of its container) and a gas (particles far apart, moving independently, fill the container), and use it to explain melting of sulphur, paracetamol and sugar from solid to liquid, and the smell of sulphur dioxide spreading as a gas. Third, they write a claim-evidence-reasoning (CER) paragraph answering: 'Why can salt and sugar not be filtered out, but sand and iron can?' with evidence from the water test and mixture M, and they distinguish physical properties (solubility, magnetism, melting point, density) from chemical properties (burning, reacting with acid). They then use the properties to answer Wanjiru's puzzle in part: 'Why does the antacid (F) fizz in vinegar but paracetamol (G) does not?' by noting that F reacts with acid while G shows no reaction, so they are different substances with different properties.	why substances that look like powders behave so differently?' Give WAIT TIME of 10-15 seconds, then cold-call 4 learners from 4 different groups. Press for evidence: 'Which result tells you that? Point to it in your table.' Model a particle diagram on the board for ice-like melting, using sulphur at about 115 C: 'Particles in the solid are tightly packed; with heat they move more and the solid becomes a liquid.' Ask: 'When salt dissolves, where do the salt particles go? Why can the filter paper not catch them?' (Particles are mixed in among the water particles and are too small to be caught.) Ask 2 learners to contrast filtration, magnetic separation and evaporation by using only these three terms. Give the CER sentence frame: 'My claim is... The evidence is... This happens because...' Correct: 'A substance that dissolves has not been destroyed. Evaporation brought the salt back, 2.7 g of it.' Tell learners to link physical properties (solubility, magnetism, melting point) and chemical properties (burning of D, acid reaction of F). Use the Kenyan hook: 'Chemists at Magadi separate soda ash from lake water in a similar way, using evaporation.' Close with: 'Properties come from what each substance is made of. That is why we can choose, separate and make materials.' Invite an alternative view: 'Does anyone disagree? Say it respectfully, using the evidence.'	the evidence to the claim. This builds understanding because the particle model explains why different substances respond differently, rather than just listing behaviours. Comparing the new groups with the Lesson 1 sorting is a revise-your-thinking routine that shows learners how evidence changes ideas.	dissolve' and of 'filtration', 'magnetic separation' and 'evaporation'; (3) particle diagrams that show solids tightly packed, liquids close but moving, gases far apart; (4) correct sorting of at least 4 properties into physical (solubility, magnetism, melting point, density) and chemical (burning, reacting with acid); (5) at least 6 of 8 groups revise their Lesson 1 sorting with an evidence-based reason. Misconception check: if a learner says 'sugar is gone when it dissolves', ask 'How could we get it back?' and note whether they suggest evaporation. Exit ticket: 'Name one property that separates D from C and one that separates F from G.'
Driving Question Board (DQB) Creation	Learners return to the class Driving Question Board started in Lesson 1, which holds the driving question: 'How do the different	Say: 'Walk to the Driving Question Board with your group. Read the questions from Lesson 1. Which ones can we now partly answer	Driving Question Board with 'We know / We wonder' notes: tracking questions publicly shows learners how evidence grows into	Indicators: (1) each group posts one evidence-based 'We know' note and one new 'We wonder' question; (2) at least 5 groups link

 VIDEO: States of matter - dubbed(KY) Source: Khan Academy (Kiswahili)  Search ARES for similar videos  HTML: States of Matter: Basics Source: PhET Interactive Simulations (English)  Search ARES for similar readings	properties of substances around us help chemists make medicines, foods and other products, and how do we use them safely?' Each group writes 2 sticky notes: one 'We now know...' note with a finding from today's evidence (for example 'Salt and sugar dissolve and cannot be filtered out; they need evaporation') and one new 'We still wonder...' question (for example 'Why does the antacid react with acid while paracetamol does not?' or 'Why is too much paracetamol dangerous even though it does not dissolve in water much?'). They place notes next to the three KICD key inquiry questions, mark Lesson 1 questions that are now partly answered with a tick, and identify questions that need later lessons.	with today's evidence?' Give WAIT TIME of 10-15 seconds before groups write. Ask: 'Which evidence, a number or an observation, supports what you are putting on the board?' Cold-call 3 groups to read their notes aloud. Sort questions together into categories: properties and separating (answered today), classifying substances (Lesson 3), medicines and safety (later lessons). Say: 'Wanjiru's puzzle about the antacid and paracetamol is still partly open. What more would we need to know?' Do not answer fully; say: 'We will return to it.' Tell learners that the DQB will be used to build model v1 in Lesson 3. Praise respectful listening: 'I saw groups acknowledging ideas from each member. That is respect in action.' Use Kenyan context if useful: 'Which of these questions would a pharmacist in Nairobi or a chemist at a Kericho tea factory ask?'	understanding and keeps the investigation anchored to the Kisumu tray and the driving question. Sorting the notes into categories shows which questions today's evidence answered and which need new evidence, so the next lessons feel purposeful rather than random.	a note to a specific result (for example '2.9 + 2.8 + 2.7 = 8.4 g' or 'sulphur burns with a blue flame'); (3) at least 3 new questions concern medicines or safety, showing movement toward the driving question; (4) learners tick at least 3 Lesson 1 questions as partly answered; (5) learners listen to each other and acknowledge ideas without interrupting. If notes only restate observations without questions, prompt: 'What does this make you wonder about medicines, food or detergents?'
Model Building Phase  VIDEO: Video: States of Matter Source: PhET Interactive Simulations (English)  Search ARES for similar videos  HTML: States of Matter Source: PhET Interactive Simulations (English)  Search ARES for similar readings	Learners begin a draft of the cumulative class model for the Kisumu tray. On a half sheet in their notebooks, each learner sketches a two-part draft: (1) a table linking each substance (A-H) to at least one property that explains its behaviour in water or on heating, and (2) a small particle picture explaining why salt dissolves and cannot be filtered while sand does not dissolve and can be filtered. Groups then share, merge the best ideas into one group sketch, and add a 'still unsure' box (for example, 'why the antacid fizzes with acid' or 'why paracetamol melts at about 169 C yet crumbles slowly in water'). The group sketches are kept in a folder	Say: 'Today we start a draft only. This is not the final model. In Lesson 3 we will build the first full class model, version 1, after we classify the substances.' Ask: 'What must your draft explain? Think about the tray, the water test, the heat test and mixture M.' Give WAIT TIME of 10-15 seconds before sketching. Circulate and ask each group: 'Which property explains why E can be pulled out and C cannot?' Cold-call 3 learners: 'Amina, how does your drawing show why salt could not be filtered?' Ask for one gap in each group's draft and write it in the 'still unsure' box. Remind learners that the draft must include the phrases 'physical property' and	Draft modelling with gaps made explicit: learners represent their current understanding in a table and a particle sketch and note what is still unexplained. Writing a 'still unsure' box makes uncertainty a normal part of science and sets up productive revision in model v1 (Lesson 3), v2 (Lesson 6) and the final model (Lesson 10). Combining a table and a particle picture links what they observe (macro) to the particle explanation.	Observable indicators: (1) each learner's table links all 8 substances to at least one property, for example A to dissolves, E to magnetism, D to burning, F to reacts with acid, G to melts at about 169 C, H to melts at about 186 C and chars; (2) particle sketches show salt particles mixed with water particles and sand particles remaining separate; (3) each group's 'still unsure' box names a specific gap; (4) exit prompts use the word 'properties'; (5) learners do not claim to have defined elements, compounds or mixtures yet, since that belongs to Lesson 3. Use a 3-point rubric: 3 = all substances linked with correct properties and a particle picture; 2

	to be used in Lesson 3 to create model v1 after classification into elements, compounds and mixtures.	'chemical property' at least once. Collect folders. Close with one exit prompt: 'Complete: Today I learned that substances behave differently because ...' Preview Lesson 3: 'We will classify A-H as elements, compounds or mixtures and use your drafts to build model v1.' Tell learners to bring one example of a food or product from home with its label for later lessons, if available.		= most linked; 1 = observations only.
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D. TEACHER REFLECTION

After the lesson, reflect on: (1) Did learners move from 'they look the same' to 'each has its own properties'? Which evidence (the water test, the heat test or the mixture M masses) persuaded them most? (2) Did every learner contribute respectfully in group discussion, including quieter learners? (3) Did the safety rules (goggles, no tasting, teacher-led heating, ventilation for sulphur) hold throughout? (4) Which misconceptions persisted, such as 'dissolved means destroyed' or 'sulphur dissolves', and how will you address them in Lesson 3? (5) Which Lesson 1 questions are still unanswered, and are they on the DQB? (6) Did learners understand why 0.6 g was lost (9 - 8.4 g) without saying it was destroyed? (7) Where could local examples, such as Magadi soda ash or sugar from Mumias and Nzoia, make the ideas stronger next time?

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	Record what you observed for each of A-H in the water test and heat test, and the masses recovered from mixture M: iron 2.9 g, sand 2.8 g, salt 2.7 g, total 8.4 g, with a loss of 0.6 g from 9 g.
What did I learn?	Write what you learned about properties of matter and states of matter: physical properties (solubility, magnetism, melting point, density) and chemical properties (burning, reacting with acid), and how solids, liquids and gases differ in their particles.
How does this explain the phenomenon?	Explain in one or two sentences, using evidence, why substances that look like powders behave differently and why salt and sugar cannot be filtered out while sand and iron can be separated by filtration or magnetic separation, and how this links to the driving question about medicines, foods and safety.

LESSON 3 (40 minutes): Matter Continued: Classifying Substances and Building Class Model v1

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	Learners use the Lesson 2 evidence on properties and separations to classify the eight Kisumu tray substances (A-H) as elements, compounds or mixtures, and use that evidence to draw the first version (v1) of the class Driving Question Board (DQB) model.
Knowledge	Matter is anything that has mass and occupies space. Element (iron E, sulphur D), compound (salt A, water B, sugar H, calcium carbonate in antacid F) and mixture (sand-iron-salt mixture M; tablets are mixtures of an active ingredient with binders). Sand C is treated as mainly silicon dioxide. Properties come from what a substance is made of. Mixtures can be separated by magnetic separation, filtration and evaporation. Recovered masses from mixture M: iron 2.9 g, sand 2.8 g, salt 2.7 g; total 8.4 g from 9 g, loss 0.6 g.
Skills	Classifying using evidence, analysing and interpreting data (mass recovery calculation), developing and using models (NGSS SEP 2), constructing explanations from evidence (NGSS SEP 6), communicating and collaborating in groups.
Attitudes	Respect: shows acceptance and appreciates the opinion of each group member while debating classifications. Curiosity and honesty in reporting evidence, including measurement loss. Safety-consciousness in handling substances (no tasting, goggles on).
Key Inquiry Question	Do all substances around the world have the same properties and uses? Why are food products, detergents and the man-made things around us the way they are?
Purpose in Storyline	Lesson 2 gave learners the WATER TEST, HEAT TEST and MIXTURE SEPARATION evidence on properties and separations. Lesson 3 asks: what are these substances made of, and can we group them by composition? It produces class model v1, which Lesson 4 uses when defining chemistry, Lesson 6 revises to v2, and Lesson 10 finalises. It moves the driving question forward by showing that a substance's properties come from what it is made of.
Safety Notes	Goggles on for all hands-on work. No tasting or smelling directly any of the substances A-H (including sugar H and the tablets). Wash hands after handling iron filings, sulphur and tablets. Any heating demonstration is teacher-led only. Keep the tablets F and G away from the mouth; they are NOT sweets. Do not give children adult doses of any medicine. Wipe up spills of water immediately to prevent slipping.

B. LESSON OVERVIEW

Mrs Achieng returns the eight Kisumu trays (A salt, B water, C sand, D sulphur, E iron filings, F antacid tablet, G paracetamol tablet, H sugar) to the eight groups. Learners predict how to group the substances by what they are made of, then examine their Lesson 2 evidence: the water test, the heat test and the mixture M separation (iron 2.9 g, sand 2.8 g, salt 2.7 g, total 8.4 g, loss 0.6 g). Using a classification card sort and evidence, learners decide which are elements (D, E), compounds (A, B, H, and calcium carbonate in F), and mixtures (M, and tablets F and G as an active ingredient with binders; sand C is treated as mainly silicon dioxide). They explain why salt dissolved in water needs evaporation, not a magnet or filtration. Finally learners draw model v1 on the class DQB, linking composition to properties to uses (medicines, foods, detergents) and safety, and add new questions for later lessons.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase	Group 1 (Otieno, Wanjiru, Amina)	Say: 'Welcome back to the Kisumu	Eliciting prior ideas (Predict-	Observable indicators: (1) every

 VIDEO: Types of mixtures Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Compounds and Mixtures Practice Source: CK-12  Search ARES for similar readings	and Kipchirchir) and the other 7 groups receive the tray of A-H. Each learner silently writes a prediction in their notebook: 'Which of A-H do you think are made of just one kind of material, and which are made of several things mixed together? Give one reason from what we saw in Lesson 2.' Then learners place sticky notes of their predicted 'pure single substance' vs 'mixed' on a two-column chart. They connect it to the phenomenon: the antacid F and paracetamol G both look like white tablets, but do they contain just one thing?	tray. In Lesson 2 you found that A to H behave differently when mixed with water and heated. Today we ask: what are they made of?' Hold up tablet F and say: 'Is this tablet one thing or many things? Write your prediction silently.' WAIT TIME 10-15 seconds, then another 10 seconds for writing. Circulate and read at least 6 predictions. Cold-call 3 learners (one from each of three different groups): 'Which substance did you place under "mixed", and what evidence from Lesson 2 made you choose it?' Do NOT confirm or correct; say: 'Thank you, we will test these ideas with evidence.' Remind: goggles on, no tasting any substance.	Observe-Explain): learners commit to a prediction before seeing new evidence, which surfaces ideas such as 'powder means pure' that can be tested and revised later. Writing before talking gives every learner a voice and builds ownership of the later revision.	learner has a written prediction and a reason with a Lesson 2 observation (e.g., 'iron was pulled out by a magnet, so it is separate'); (2) teacher tallies how many sticky notes place the antacid F and paracetamol G under 'pure' vs 'mixed'; (3) the 3 cold-called learners refer to evidence, not just opinion. Note misconceptions such as 'white powder = one substance' or 'dissolved means gone'.
Observe Phase  VIDEO: Suspensions, colloids and solutions - dubbed(KY) Source: Khan Academy (Kiswahili)  Search ARES for similar videos  EXERCISE: Compounds and Mixtures Practice Source: CK-12  Search ARES for similar readings	Learners review their own Lesson 2 evidence and run a short, safe hands-on check. Task 1: using the recorded WATER TEST and HEAT TEST results for A-H, groups complete a data table: dissolves or not, magnet response, burns or melts. Task 2: using mixture M (3 g iron filings + 3 g sand + 3 g salt, total 9 g), groups re-trace the separation steps: magnetic separation removes iron (recovered 2.9 g), filtration removes sand (2.8 g after drying), evaporation of the filtrate recovers salt (2.7 g). Learners calculate $2.9 + 2.8 + 2.7 = 8.4$ g and the loss $9 - 8.4 = 0.6$ g and note it is from spillage and sticking to equipment. Task 3: learners note that salt dissolved in water passes through the filter paper and cannot be removed by a magnet, so it needs evaporation. They observe that each part of M kept its own properties (iron stayed magnetic,	Say: 'Open your Lesson 2 data. Today you are chemists checking the evidence. Which substances stayed the same in water? Which changed?' Distribute the data table and wipe-board marker. Prompt: 'Look at mixture M. After separating it, did iron, sand and salt each keep its own properties? Show me the evidence in your table.' WAIT TIME 10-15 seconds after each question. Cold-call 4 learners across different groups: 'Why could we not use filtration to get salt out of the water?' and 'Where did the missing 0.6 g go?' Walk to Group 1 and ask Amina: 'What does the 8.4 g tell you about how careful the group was?' Reinforce safety: 'Goggles stay on; the heat results are from the teacher-led demonstration, so no one heats anything today. Do not taste or sniff any substance.' If a group is stuck, ask: 'Which separation method uses a physical	Data analysis and pattern finding (NGSS SEP 4: Analysing and interpreting data): learners tabulate properties and calculate recovered mass to look for patterns, such as 'substances in a mixture keep their own properties and can be separated by physical means'. Using numbers (8.4 g, loss 0.6 g) trains evidence-based, honest reporting.	Observable indicators: (1) data table complete for all 8 substances A-H with correct entries (e.g., E pulled by magnet, A and H dissolve, D burns blue); (2) correct arithmetic $2.9 + 2.8 + 2.7 = 8.4$ g and loss 0.6 g; (3) learners state that salt cannot be filtered out because it dissolves and needs evaporation; (4) learners use the exact terms 'filtration', 'magnetic separation' and 'evaporation'. Check with a thumbs signal and note groups needing support.

	sand stayed gritty and undissolved, salt still tasted... NOT tested, but dissolves) so M is a mixture. Learners also note the key contrast: sulphur D burns with a blue flame and a sharp choking smell (sulphur dioxide) while sand C shows no change on heating; sugar H melts at about 186 C, browns and chars with a caramel smell, while salt A just crackles.	property such as magnetism, size of particle, or dissolving?		
Explain Phase  VIDEO: 01 Pure Substances Vs Mixtures Source: Seavuria  Search ARES for similar videos  EXERCISE: Separation of solutions and mixtures chromatography <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum)  Search ARES for similar readings	Groups use a classification card sort. Cards show A salt, B water, C sand, D sulphur, E iron filings, F antacid tablet, G paracetamol tablet, H sugar and mixture M. Groups place each under Element, Compound or Mixture using the class definitions: element (iron E, sulphur D), compound (salt A, water B, sugar H, and calcium carbonate in F), mixture (M; tablets F and G are mixtures of an active ingredient with binders). Sand C is treated as mainly silicon dioxide. Learners write a claim-evidence-reasoning (CER) statement for two substances, for example: 'Claim: E iron filings is an element. Evidence: ... Reasoning: ...'. Learners answer Wanjiru's puzzle: 'Why does the antacid relieve heartburn while paracetamol relieves pain?' They explain that each tablet has its own active ingredient with its own properties, so the antacid reacts with and neutralises acid (it fizzes in dilute vinegar or lemon juice, while G shows no fizz), whereas paracetamol works on pain and fever by a different action. They also state that both are harmful in excess: adult paracetamol label 1 to 2 tablets of 500 mg every 4 to 6 hours, maximum 8 tablets (8 x 500 mg = 4000 mg = 4 g) in 24 hours;	Say: 'Chemists classify substances so they can predict how they behave. An element is one kind of substance that cannot be split into simpler ones by ordinary methods, like iron and sulphur. A compound has different elements joined together, like salt, water and sugar. A mixture has substances just mixed, each keeping its own properties.' Then: 'Use the cards and decide, with evidence.' WAIT TIME 10-15 seconds before groups begin. Rotate between groups and ask: 'Respect each member's idea; Kipchirchir, what do you think and what does the group say?' Cold-call 4 groups to defend one classification. Ask: 'Why is the antacid tablet F a mixture even though it contains a compound, calcium carbonate?' Then ask: 'Wanjiru asked why the antacid and paracetamol differ. Using the fizz-in-vinegar evidence, who can explain?' WAIT TIME 10-15 seconds, then take 3 responses. State the safety link: 'Never give children adult doses; always follow the label or a health worker's advice; check the Pharmacy and Poisons Board approved medicines.' Do not reteach the definition of chemistry or the branches; those come in Lessons 4 and 5.	Claim-Evidence-Reasoning with card sort (NGSS SEP 6: Constructing explanations): learners must justify each classification with evidence from their own A-H results, linking composition to properties. Structured peer debate with equal voice practises the value of respect and exposes where ideas differ.	Observable indicators: (1) card sort correct: D and E elements; A, B, H compounds; M, F and G mixtures; C treated as mainly silicon dioxide; (2) each CER statement cites at least one specific observation (e.g., 'F fizzes in vinegar, G does not'); (3) learners explain that the antacid neutralises acid while paracetamol relieves pain and fever and does not neutralise acid; (4) learners quote 4 g per 24 hours and the liver risk. Use a 3-2-1 check: three classifications justified, two properties linked, one safety rule. Record learners who respect others' views during the debate.

	more can damage the liver; antacid taken too often can cause constipation or diarrhoea.			
Driving Question Board (DQB) Creation  VIDEO: Types of mixtures Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Chemistry of Compounds Practice <i>Related topic: not an exact match for this lesson.</i> Source: CK-12  Search ARES for similar readings	Groups return to the class DQB, which began in Lesson 1. They sort existing question cards into categories: 'What are things made of?', 'How can we separate them?', 'How do medicines work and stay safe?'. They mark questions they can now partly answer (for example, 'Why can some mixtures be separated easily while others cannot?') and write the answer with evidence on a green card beside it. They add at least one new question per group, for example 'How do chemists make a medicine for chronic diseases from these materials?', 'Why does the sugar from Mumias or Nzoia dissolve but sand does not?' and 'Do all substances around the world have the same properties and uses?'. Group 1 links the 0.6 g loss to a question about how factories in Kenya, such as Magadi soda ash, avoid wasting material.	Say: 'Look at our Driving Question: how do the different properties of substances help chemists make medicines, foods and other products, and how do we use them safely? Which question cards can we now answer, even partly?' WAIT TIME 10-15 seconds. Invite 3 learners to place a green evidence card and read it aloud. Cold-call 2 groups to share their new question and say which KICD key inquiry question it links to. Ask: 'Which questions are still open? Where will we look for the answers?' Star at least 2 open questions for Lessons 4 to 9 (for example, the meaning of chemistry, branches, industries, labels and drug use). Praise respectful listening: 'Thank you for acknowledging each other's ideas.'	Driving Question Board as a public record of growing understanding: separating answered, partly answered and open questions makes progress visible and keeps learners' curiosity steering the unit. Connecting each card to evidence reinforces that scientific claims need data.	Observable indicators: (1) each group posts at least one evidence-backed answer card and one new question; (2) at least 2 new questions connect composition or properties to medicines, foods or detergents; (3) learners can name a KICD key inquiry question that matches their card; (4) teacher notes which questions show persistent misconceptions (for example, 'dissolved salt disappears').
Model Building Phase  VIDEO: Suspensions, colloids and solutions - dubbed(KY) Source: Khan Academy (Kiswahili)  Search ARES for similar videos  EXERCISE: Compounds and Mixtures Practice Source: CK-12  Search ARES for similar	Each group draws model v1 on a chart paper for the class DQB model. The model shows: Matter (anything with mass that occupies space) at the top; three boxes: element (D sulphur, E iron), compound (A salt, B water, H sugar, calcium carbonate in F) and mixture (M, F, G); arrows from composition to properties (solubility, magnetism, burning, melting, reaction with acid) and then to uses and separations: magnetic separation, filtration and evaporation. A section labelled 'Medicines and safety' shows the antacid neutralises acid, paracetamol relieves pain and	Say: 'A model is a tool for explaining and predicting. Draw what you think explains why A to H behave differently and how that helps chemists make medicines, foods and detergents safely.' Show the required parts: matter, three classes, properties, uses and safety. WAIT TIME 10-15 seconds before groups start drawing. Circulate and ask: 'Which piece of evidence supports this arrow? What would you change if mixture M had lost 2 g instead of 0.6 g?' Hold a gallery walk of 5 minutes; learners leave one sticky note with a 'star' (what is strong) and one 'wish' (what is missing). Cold-call 3	Developing and using models (NGSS SEP 2): learners represent relationships among composition, properties, separations and safe use in one diagram. Peer gallery walk with star and wish feedback develops critique skills, and labelling uncertainty teaches that models are revised as new evidence arrives.	Observable indicators: (1) model v1 includes matter, the three classes with correct examples, properties, separation methods and the medicine-safety link; (2) at least two arrows are supported with a specific piece of evidence; (3) a 'not sure yet' box is present; (4) the exit ticket correctly says that iron and sulphur are each a single kind of substance (elements), while salt is a compound. Collect exit tickets and sort into secure, partly secure and needs support.

readings	fever, and the 4 g per 24 hours maximum with the liver warning. Groups add a 'We are not sure yet' box. Groups present in a gallery walk and one representative from each group explains one arrow using evidence. The class agrees on one shared v1 and keeps it for revision in Lesson 6 (v2) and Lesson 10 (final).	groups to explain one arrow. Then combine into one class v1 on the DQB wall and say: 'This is version 1; it is allowed to be incomplete. We will revise it in Lesson 6 and finalise it in Lesson 10.' Close with an exit ticket: 'Write one sentence explaining why iron and sulphur are elements while salt is a compound, using evidence.' Remind learners to wash their hands and leave the trays tidy.		
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D. TEACHER REFLECTION

Which learners still believe that a white powder or a tablet is automatically a single substance, and what evidence changed (or did not change) their minds? Did every group member, including quieter learners such as those in Group 1, contribute to the card sort and the model with respect shown to others' ideas? Were the 0.6 g loss and the 4 g per 24 hours limit understood as real data rather than numbers to memorise? Which model v1 gaps (for example, why elements and compounds differ at a deeper level) should be noted on the DQB for later lessons, and how will Lesson 4 build on this evidence without re-teaching it?

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	We observed that A salt and H sugar dissolve completely, C sand and E iron filings do not dissolve, E is pulled out by a magnet, D sulphur floats and burns with a blue flame and sharp smell, and F antacid fizzes in vinegar while G paracetamol does not. In mixture M we recovered iron 2.9 g, sand 2.8 g and salt 2.7 g, a total of 8.4 g out of 9 g, so 0.6 g was lost.
What did I learn?	We learned that substances can be classified as elements (D sulphur, E iron), compounds (A salt, B water, H sugar, calcium carbonate in F) or mixtures (M, and tablets F and G with binders). Mixtures can be separated by magnetic separation, filtration and evaporation, but dissolved salt needs evaporation. Class model v1 links composition to properties to uses and safety.
How does this explain the phenomenon?	The substances behave differently because each has its own properties, which come from what it is made of. Iron and sand can be pulled out or filtered, while salt and sugar dissolve and cannot be filtered out. The antacid works because it neutralises acid, paracetamol works on pain and fever by a different action, and both are harmful in excess, so we follow labels and correct dosage.

LESSON 4 (40 minutes): Meaning of Chemistry as a Field of Science: Defining Chemistry from the Tray Evidence

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	Learners explain the meaning of Chemistry as a field of science by building a definition from the Kisumu tray evidence (substances A-H) and testing it against their own observations.
Knowledge	Chemistry is the science that studies matter: its composition, properties, changes and uses. Matter is anything that has mass and occupies space, and exists as solid, liquid or gas. Each tray substance has its own properties (solubility, magnetism, melting and burning behaviour, reaction with acid) that come from what it is made of. Chemists use these properties to choose, separate and make materials such as medicines, foods and detergents.
Skills	Constructing explanations from evidence; communicating and collaborating in pairs and groups (KICD core competency: contributing to group decision making by recognising and acknowledging the value of others' ideas); analysing and interpreting data from the tray observations; developing and refining a definition (NGSS SEP: Obtaining, Evaluating and Communicating Information; Constructing Explanations).
Attitudes	Respect: learners show acceptance while appreciating the opinion of each member as they discuss and refine the definition. Curiosity about how chemists use properties of substances to make safe products.
Key Inquiry Question	KICD key inquiry questions: (1) How are medicines for chronic diseases developed? (2) Do all substances around the world have the same properties and uses? (3) Why are food products, detergents and the things in the surrounding, especially those that are man-made, the way they are? This lesson answers the first part of each by asking: What is Chemistry, and what in our tray evidence shows it is the science of matter?
Purpose in Storyline	Lessons 1 to 3 gave learners the Kisumu tray observations, the properties and separation evidence, the classification of substances and class model v1. Lesson 4 gives learners the language to name what they have been doing: Chemistry. The class definition becomes the foundation for Lesson 5 (the five branches) and later lessons on industries, careers, labels and drugs, and it advances the driving question by showing that knowing properties is the heart of how chemists make products and keep people safe.
Safety Notes	No new practical work is done. If learners handle tray substances for re-observation, goggles must be worn, nothing is tasted or put near the mouth (including the antacid F, paracetamol G and sugar H), hands are washed after handling, and heating is done only by Mrs Achieng or under her direct supervision. Paracetamol and antacid tablets are used for observation only and are never swallowed in class. Remind learners that medicines are taken only as advised on the label or by a health worker.

B. LESSON OVERVIEW

Mrs Achieng returns the groups to their Kisumu trays (A table salt, B water, C sand, D sulphur, E iron filings, F antacid tablet, G paracetamol tablet, H sugar). Learners first predict what Chemistry means and what chemists study, then gather evidence by sorting their own tray observations into the categories composition, properties, changes and uses. In the Explain phase they build a class definition, test it against specific evidence (salt and sugar dissolving, sand and iron not dissolving, sulphur burning, paracetamol melting at about 169 C, the antacid fizzing in vinegar) and apply it to the medicine puzzle raised by Wanjiru. They then add new questions to the Driving Question Board and revise the definition section of model v1 so it names Chemistry as the science that studies matter.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase  VIDEO: Intro to Chemistry Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: Defining Chemistry (Flexbook) Source: CK-12  Search ARES for similar readings	Learners sit in their groups of four (for example Otieno, Wanjiru, Amina and Kipchirchir in Group 1) beside the tray of substances A-H. Individually, for 2 minutes, each writes in the notebook: 'I think Chemistry is the study of _____ because when we tested the tray, we _____.' They also list two things they believe a chemist in Kisumu, Mombasa or Nairobi does. Then they share with a partner and the pair agrees on one prediction sentence.	Say: 'Last lesson we classified A to H as elements, compounds and mixtures and built model v1. Today we name the science we have been doing. Before I tell you anything, write what you think Chemistry is.' Display the DQB and the tray. After the question, give WAIT TIME of 10-15 seconds of silent thinking before anyone writes. Walk around and note recurring words (substances, mixing, medicine). Cold-call 4 learners from different groups to read their pair's sentence without comment. Say: 'Thank you, I am not marking right or wrong yet. We will test these ideas against our evidence.' Record 4 predictions on the board under the heading 'Our first ideas about Chemistry'. Remind the class that all ideas are valued and respectful listening is expected.	Predict-Share-Revisit: learners commit to an idea before instruction so their later revision is visible to themselves. Writing the sentence first, then sharing with a partner, surfaces prior ideas, including misconceptions such as 'Chemistry is only about dangerous liquids and laboratories', and creates a baseline to compare against the final definition.	Observe whether predictions mention substances, properties, medicines or changes (versus only 'experiments'). Indicators: at least 4 of 8 groups include 'substances' or 'materials' in their sentence; note learners who link to a tray result (for example 'salt vanished in water'). Tally misconceptions (Chemistry = only chemicals that are dangerous; Chemistry = only lab work) on a teacher checklist to address in the Explain phase.
Observe Phase  VIDEO: Intro to Chemistry Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Defining Chemistry Practice Source: CK-12  Search ARES for similar readings	Learners review their own recorded observations from Lessons 1-3 (water test, heat test, mixture separation, classification) and the tray in front of them. Each group receives a four-column evidence chart headed Composition (what it is made of), Properties, Changes, Uses. Using their data, they fill in at least one row per substance for six chosen substances (A to H, minimum six). Examples: A salt dissolves completely in water and crackles but does not melt when heated; C sand sinks, stays visible and does not change when heated; D sulphur floats, melts at about 115 C then burns with a blue flame and	Say: 'You are chemists today. Sort what you already know about A to H into four columns. Use only evidence from your own results.' Model one row on the board (A table salt: composition: sodium chloride, a compound; property: dissolves completely, clear colourless solution; change: crackles but no melting when heated; use: seasoning food, for example in ugali and nyama choma). Circulate to every group at least once and ask: 'Which column does the blue flame of sulphur belong in, and why?' and 'Where does the magnet separating E from the mixture M belong?' Give WAIT TIME of 10-15	Evidence Sorting with a Four-Column Chart (Composition, Properties, Changes, Uses): learners organise familiar data under the categories that will become the parts of the definition. This turns scattered observations into patterns (every substance has a composition and properties, many show changes, and uses follow from properties) and prepares them to build the definition from evidence rather than memorising it.	Check charts for accuracy against the established results: salt and sugar listed as dissolving, sand, iron and sulphur as not dissolving; sulphur burning blue; paracetamol melting at about 169 C; sugar melting at about 186 C. Indicator: at least 6 substances entered with at least three of the four columns filled. Watch for property/change confusion (for example listing 'burns' as a property of sand) and for correct use of letters A-H. Note which groups can link a use (such as antacid for heartburn) to a property (reacts with and neutralises acid).

	sharp choking smell (sulphur dioxide); E iron filings are pulled out by a magnet and glow orange-red then turn dull blackish when heated; F antacid crumbles into a cloudy white suspension and fizzes in dilute vinegar; G paracetamol stays mostly intact in water, does not fizz, melts at about 169 C and browns; H sugar dissolves, melts at about 186 C, turns brown then chars black with a caramel smell. They also use mixture M: 9 g in, 8.4 g recovered (2.9 g iron, 2.8 g sand, 2.7 g salt), loss 0.6 g. Learners may briefly handle the tray substances with goggles on and no tasting to re-check a result.	seconds after each question before prompting. Check safety: goggles on, no tasting, tablets not swallowed. Cold-call 3 groups to read one row aloud and ask the class to point to the evidence in their own notes. Praise respectful use of others' ideas, for example 'I like how Amina asked Kipchirchir to explain his row before adding hers.'		
Explain Phase  VIDEO: Intro to Chemistry Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: Chemical Properties of Matter (Flexbook) Source: CK-12  Search ARES for similar readings	1) Think-Pair-Share: pairs scan their chart and write answers to 'What do all four columns have in common?' (they all describe matter, anything that has mass and occupies space). 2) Groups use sentence frames to draft a definition: 'Chemistry is the science that studies _____, including its _____, _____, and _____.' 3) Groups present drafts to the class; the class compares and negotiates one agreed class definition: Chemistry is the science that studies matter: its composition, properties, changes and uses. 4) Learners test the definition against three challenge cases: (a) Does it explain why salt dissolves but sand does not? (b) Does it explain how the mixture M was separated by a magnet, filtration and evaporation? (c) Wanjiru's question: why does the antacid F relieve heartburn while paracetamol G relieves pain? They explain that the antacid reacts with	Say: 'Look at your four columns. What is the same about every substance on this tray?' Wait 10-15 seconds, then cold-call 3 learners. Introduce the term matter: 'Matter is anything that has mass and occupies space. Water, sand, a tablet and the air in this room are all matter.' Briefly remind learners of solids, liquids and gases without re-teaching Lessons 2 and 3. Give groups 4 minutes to draft the definition. Select 3 groups to present (different wording, for example one group that stresses uses). Write the common words on the board and negotiate: 'Which words do all three drafts share? What is still missing?' Then give the conventions definition and have learners compare it to the class version. For challenge case (c), say: 'Wanjiru asked this in Lesson 1. Use the definition to explain what you now understand.' Give WAIT TIME of 10-15 seconds, then cold-call 4 learners across groups. Stress the safety	Claim-Evidence-Reasoning with Consensus Discourse: learners make a claim (the definition), support it with specific tray evidence, and reason why the evidence supports each part of it. Negotiating a class definition from several group drafts builds shared ownership, and testing the definition on challenge cases (including the medicine puzzle) shows that a good definition can explain real phenomena, not just be recited.	Look for definitions that include matter plus at least three of composition, properties, changes and uses. Indicators: learners cite specific evidence (for example 'Sulphur D melts at about 115 C then burns, which is a change, so Chemistry studies changes'); learners correctly explain that the antacid neutralises acid and paracetamol does not; learners state the 4 g maximum correctly; learners acknowledge a peer's idea respectfully during negotiation. Use a quick 3-finger check (3 = confident to explain, 1 = need help) and revisit with groups showing 1.

	and neutralises excess stomach acid (it fizzes in vinegar) while paracetamol relieves pain and fever by a different action and does not neutralise acid. 5) Learners state why taking too much of either is dangerous: paracetamol maximum 8 tablets x 500 mg = 4000 mg = 4 g in 24 hours, more can damage the liver; too much antacid can upset the body's salts and cause constipation or diarrhoea. Each learner finishes by writing the agreed definition in their own words and linking it to one piece of tray evidence.	message: 'Chemistry also helps us recognise the risks of drug abuse and develop safe medicines. Always follow the label or a health worker; never give children adult doses.' Address misconceptions noted earlier (Chemistry is not only dangerous liquids; it studies all matter, including ugali, tea from Kericho and the water in Lake Victoria). Do not teach the five branches here; say: 'Next lesson we will see that Chemistry has branches.'		
Driving Question Board (DQB) Creation  VIDEO: Intro to Chemistry Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Chemical Properties of Matter Practice Source: CK-12  Search ARES for similar readings	Learners return to the class Driving Question Board. In groups, they discuss: 'Which of our earlier questions can we now answer using the definition of Chemistry?' They place sticky notes on answered questions with the evidence (for example 'Why do powders behave differently? Because each has its own composition and properties'). Each group then writes at least one new question that this lesson raises, such as 'What are the different areas of Chemistry?', 'Who works as a chemist in Kenya?', 'How do chemists in Nairobi make sure medicines are safe?' or 'Why does the label on paracetamol give a maximum number of tablets?' and adds it to the board under 'Questions to answer next'.	Say: 'Our driving question asks how properties help chemists make medicines, foods and other products, and how we use them safely. Which of our questions can we now answer?' Wait 10-15 seconds. Cold-call 3 groups to move one answered question and read the evidence. Cold-call 2 more groups for new questions and ask: 'What in today's lesson made you ask that?' Group new questions into likely lessons: branches (Lesson 5), industries (Lesson 6), careers (Lesson 7), labels (Lesson 8), drugs and substance abuse (Lesson 9). Say: 'We will return to these as we gather more evidence.' Ensure that every group contributes at least one note and that contributions are acknowledged respectfully.	Driving Question Board as a Public Record of Learning: the board makes visible what the class has figured out and what remains puzzling. Linking questions to evidence turns answered questions into claims, and new questions created from the definition show learners that understanding grows in steps and guides the next lessons.	Check that each group adds at least one new question and moves at least one answered question with evidence. Indicators: new questions refer to the definition or the tray (not generic questions); answered questions cite A-H data. Note groups whose questions are only recall (for example 'What is Chemistry?') and prompt them to ask 'how' or 'why' questions linked to medicines, foods or detergents.
Model Building Phase  VIDEO: Intro to Chemistry Source: Khan Academy (English	Learners revisit class model v1 (from Lesson 3), which classified A-H as elements, compounds and mixtures. In groups they add a header or box to the model: 'Chemistry = the science that studies matter: composition,	Say: 'Our model v1 shows how we classify the substances. Now add what the whole science is about.' Display model v1 and model one addition (for example an arrow from A salt to 'properties: dissolves in water'). Circulate and ask:	Cumulative Model Revision: learners add the new idea (the definition of Chemistry) to the existing model v1 and connect it to earlier evidence. Drawing arrows between a general definition and specific data shows how big ideas	Check the models for at least three correct links between tray evidence and parts of the definition and for the use of letters A-H. Exit ticket indicators: includes 'matter' and at least three of composition, properties, changes, uses; gives

- US curriculum) Search ARES for similar videos HTML: Defining Chemistry (Flexbook) Source: CK-12 Search ARES for similar readings	properties, changes and uses.' They draw arrows linking at least three tray examples to the parts of the definition: composition (D sulphur and E iron are elements; A salt, B water and H sugar are compounds; mixture M is a mixture), properties (salt dissolves, iron is magnetic), changes (sulphur burns, sugar chars) and uses (antacid for heartburn, paracetamol for pain and fever). Each group writes one sentence on how the definition helps explain Wanjiru's question about the antacid and paracetamol. They add an arrow labelled 'safe use: label and dosage' pointing to the medicines. A volunteer from each of 2 groups explains their addition to the class.	'Which part of the definition does this arrow show?' and 'What evidence supports it?' Give WAIT TIME of 10-15 seconds after each question. Cold-call 2 groups to share. Say: 'This is not the final model. In Lesson 6 we will revise it to version 2 after we study how Chemistry is used in industries such as Magadi soda ash, Kericho tea and Mumias sugar.' Close with a 1-minute exit ticket: 'Write the class definition of Chemistry and give one piece of evidence from tray substances A-H.' Collect the tickets.	are supported by evidence and prepares for later revisions (v2 in Lesson 6 and the final model in Lesson 10).	one accurate piece of evidence (for example F fizzes in vinegar, so the antacid reacts with acid). Sort tickets into secure, partly secure and not yet secure, and plan a short revisit at the start of Lesson 5.
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D. TEACHER REFLECTION

After the lesson, consider: Did learners build the definition from evidence in their own words, or did they wait for the teacher's version? Which tray observations did learners use most readily as evidence, and which were harder to place in the four columns (composition vs properties vs changes)? Did any learners still think Chemistry is only about dangerous liquids or laboratory work? Were all voices heard in group discussion, and did learners show respect when negotiating the class definition? Did the medicine puzzle (antacid, paracetamol and the 4 g maximum) help learners see why Chemistry matters for safety? What adjustments are needed before Lesson 5, when learners match the five branches to parts of the phenomenon?

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	What did you observe about the tray substances A-H that shows Chemistry studies matter? For example: salt A and sugar H dissolve, sand C and iron E do not, sulphur D burns with a blue flame, and the antacid F fizzes in vinegar while paracetamol G does not.
What did I learn?	What did you learn about the meaning of Chemistry? Chemistry is the science that studies matter: its composition, properties, changes and uses. Matter is anything that has mass and occupies space.
How does this explain the phenomenon?	How does the definition of Chemistry help explain why the antacid relieves heartburn while paracetamol relieves pain, and why chemists in Kenya need to know the properties of substances to make medicines, foods and detergents safely?

LESSON 5 (40 minutes (one lesson)): Branches of Chemistry: Matching the Five Branches to the Kisumu Tray

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	Learners explore the five branches of chemistry (organic, inorganic, physical, analytical and biochemistry) and match each to part of the Kisumu tray phenomenon, while showing respect for each member's opinion in group discussion.
Knowledge	Learners will be able to name the five branches of chemistry and say what each studies. They will match sugar (H) and paracetamol (G) to organic, salt (A) and iron filings (E) to inorganic, melting points (sulphur about 115 C, paracetamol about 169 C, sugar about 186 C) to physical, the separation of mixture M to analytical, and how the antacid (F) and paracetamol (G) act in the body to biochemistry.
Skills	Discussing in groups, matching evidence to categories, justifying a claim with evidence from the tray (Constructing Explanations), communicating ideas respectfully, and using the Lesson 4 definition of chemistry to reason.
Attitudes	Respect: the learner shows acceptance while appreciating the opinion of each member as they discuss in groups the various branches of Chemistry. Learners also develop curiosity about how chemists work in Kenya.
Key Inquiry Question	Do all substances around the world have the same properties and uses? How are medicines for chronic diseases developed? (Learners start from the tray and ask which branch of chemistry would answer each puzzling question.)
Purpose in Storyline	Lesson 4 defined chemistry as the science that studies matter: its composition, properties, changes and uses. Lesson 5 shows that this science has five branches, each answering a different question about the tray. This gives learners a vocabulary to sort their DQB questions and prepares them for Lesson 6, where they study how chemistry is used in Kenyan industries and revise the class model to v2.
Safety Notes	No tasting or smelling any substance. Goggles on if any tray substance is handled. Tablets (F and G) are looked at only and are not eaten; they must not be taken from the lab. Sulphur (D) and iron filings (E) are handled only with spatulas. No heating in this lesson. Use data from earlier lessons and the lesson cards, not new demonstrations. Respectful talk rules apply: one speaker at a time, and build on or politely challenge ideas.

B. LESSON OVERVIEW

Mrs Achieng returns to the Kisumu tray (A salt, B water, C sand, D sulphur, E iron filings, F antacid tablet, G paracetamol tablet, H sugar). Learners recall the Lesson 4 definition of chemistry and predict what 'branches' of chemistry might mean. In Group 1 (Otieno, Wanjiru, Amina and Kipchirchir) and the other seven groups, they rotate through five evidence cards, one per branch, using the tray results already collected: sugar and paracetamol, salt and iron, melting points, separation of mixture M (recovered 2.9 g, 2.8 g and 2.7 g; loss 0.6 g), and the action of the antacid and paracetamol. Groups match each card to a branch, justify the match, and practise respect through talk norms and a listening check. The class then sorts DQB questions by branch and adds the five branches to the model notes ahead of v2 in Lesson 6.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase  VIDEO: No closely	Learners look at the tray (A-H) and the Lesson 4 definition on the board: chemistry is the science	Say: 'Last lesson we defined chemistry. Today we ask: does every chemist study the same	Eliciting prior knowledge through a Predict-Write-Pair-Share routine: learners commit ideas in writing	Look for: (1) at least 80% of learners writing a question or prediction within 2 minutes; (2)

matching video in the ARES library for this activity. Search ARES for videos Search terms: chemistry branches organic inorganic  HTML: Organic Chemistry (Flexbook) Related topic: not an exact match for this lesson. Source: CK-12 Search ARES for similar readings	that studies matter: its composition, properties, changes and uses. Individually they write in two minutes: 'Chemistry is one science, but chemists do different jobs. Which different questions about our tray might different chemists ask?' They then share with a partner, and groups write one prediction of what 'branches of chemistry' means and put a ? next to any word they are unsure of.	thing about this tray?' Point to A-H. Ask learners to write silently for 2 minutes. Then pose: 'Which question about the tray would YOU like a chemist to answer first?' WAIT TIME 10-15 seconds before taking any answer. Cold-call 4 learners (including at least one from the back row) and write their questions on the board without evaluating. Say: 'Thank you. I will not tell you if this is right yet. We will test it with evidence.' Remind the class of the respect norm: 'In this lesson, every opinion is heard before we decide.'	before discussion. This surfaces what they already think about chemistry and makes later revision of ideas visible.	predictions that mention properties, medicines, separation or food; (3) learners using the words 'matter', 'properties' or 'substances' from Lesson 4. Note learners who write only 'I do not know' for targeted support during group work.
Observe Phase  VIDEO: No closely matching video in the ARES library for this activity. Search ARES for videos Search terms: chemistry branches organic inorganic  EXERCISE: Organic Chemistry Practice Related topic: not an exact match for this lesson. Source: CK-12 Search ARES for similar readings	In groups of four, learners rotate through five evidence cards placed at stations on the lab benches, spending about 4 minutes at each. Each card gives data already collected from the tray: Card 1: H sugar and G paracetamol (made of carbon-containing substances; sugar melts at about 186 C, turns brown, then chars black with a caramel smell; paracetamol melts at about 169 C and browns). Card 2: A salt and E iron filings (a salt dissolves completely; iron is pulled out by a magnet and glows orange-red then turns dull blackish when heated). Card 3: Melting points: D sulphur melts at about 115 C, G paracetamol at about 169 C, H sugar at about 186 C; A salt crackles without melting and C sand does not change. Card 4: Mixture M (3 g iron + 3 g sand + 3 g salt = 9 g): magnet recovers 2.9 g, filtration recovers 2.8 g, evaporation recovers 2.7 g; total 8.4 g, loss 0.6 g. Card 5: F antacid neutralises excess stomach acid	Before rotation, display the five branch names with one-line meanings: organic (substances based on carbon, such as sugar and many medicines), inorganic (substances such as salts, metals and minerals), physical (how matter behaves, such as melting points and energy changes), analytical (finding out what is present and separating or measuring it), biochemistry (chemistry of living things, including how medicines act in the body). Say: 'Read the card, then ask your group: what is this card measuring or explaining?' Circulate and ask each group: 'Which words on the card made you think of that branch?' WAIT TIME 10-15 seconds after each question. Cold-call 2 recorders per round for one-sentence reports. Do not give the matches; if a group is stuck, say: 'Look at the verb on the card: separate, melt, act in the body, contain carbon, element or salt.' Safety: reinforce no tasting	Evidence-based categorising (Analysing and Interpreting Data): learners sort real tray data into categories and justify each match with a specific piece of evidence. Rotating stations spreads the data across the room and gives every learner a defined role, which supports respectful sharing.	Observable indicators: (1) recorders write the branch name AND one data point (for example '169 C' or '2.9 g') for each card; (2) in each group every member speaks at least once at each station; (3) the respect-checker marks one example of a group member building on another's idea. Note groups that assign branches using only the substance name without data.

	and fizzes in dilute vinegar; G paracetamol relieves pain and fever, does not fizz in vinegar and does not neutralise acid; too much of either is harmful (maximum 8 x 500 mg = 4 g in 24 hours for adult paracetamol). At each station, learners record what the card shows and a first guess at which branch it belongs to. Each learner takes a role: reader, recorder, timekeeper or respect-checker (Otieno, Wanjiru, Amina and Kipchirchir rotate roles in Group 1).	and no handling tablets beyond looking.		
Explain Phase  VIDEO: No closely matching video in the ARES library for this activity.  Search ARES for videos Search terms: chemistry branches organic inorganic  PDF: Organic Chemistry 1&2 Related topic: not an exact match for this lesson. Source: ARES Local Curriculum - Digital Literacy Content  Search ARES for similar readings	Groups return to their desks and complete a matching table: Branch Part of the Kisumu phenomenon Evidence Why this branch? Each group must explain one disagreement they had and how they resolved it respectfully. Then groups face the puzzle: 'Wanjiru asked why the antacid relieves heartburn but paracetamol relieves pain. Which branches help answer that question?' Learners argue that biochemistry explains how each acts in the body, inorganic or physical chemistry helps explain why the antacid neutralises acid, and organic chemistry helps explain what paracetamol is made of. They also note that one tray substance can belong to more than one branch (for example paracetamol G is studied by organic chemistry for its makeup and by biochemistry for its action). Groups share a claim, evidence and reasoning to the class.	Say: 'Choose the branch you think fits best, and tell us the evidence. Remember: we listen first, then we agree or politely disagree.' Model a respectful sentence frame on the board: 'I understand your idea that _____. My evidence suggests _____.' Cold-call 3 different groups to give one match and defend it. WAIT TIME 10-15 seconds before asking another group to respond. Ask: 'Can one substance belong to two branches? Which one, and why?' Ask Group 1 to explain how Wanjiru's puzzling question connects to biochemistry. Correct misconceptions: that each substance belongs to only one branch; that 'organic' only means farming without chemicals (here it means carbon-based substances). Confirm expected matches without over-teaching: sugar and paracetamol (organic), salt and iron (inorganic), melting points (physical), separation of mixture M (analytical), antacid and paracetamol action (biochemistry). Do not re-teach the classification of elements, compounds and mixtures (Lesson 3) or the definition of chemistry (Lesson 4); only refer to them.	Claim-Evidence-Reasoning (Constructing Explanations) with a respectful discussion protocol: learners state a claim, support it with tray data, and explain the reasoning linking the data to the branch. Using sentence frames for agreeing and disagreeing makes acceptance of others' opinions an explicit part of the thinking.	Observable indicators: (1) at least 6 of 8 groups correctly match all five branches with at least one piece of data each; (2) learners can state that paracetamol G fits more than one branch; (3) learners use the sentence frame when responding to another group; (4) correct statement that mixture M separation (magnet 2.9 g, filtration 2.8 g, evaporation 2.7 g) is analytical chemistry. Probe any group that mixes up physical and analytical by asking: 'Is this about how matter behaves, or about finding out and separating what is in it?'

Driving Question Board (DQB) Creation  VIDEO: No closely matching video in the ARES library for this activity. Search ARES for videos Search terms: chemistry branches organic inorganic  HTML: Organic Chemistry (Flexbook) Related topic: not an exact match for this lesson. Source: CK-12 Search ARES for similar readings	Learners look at the class DQB with questions from Lessons 1-4. In groups they place a sticky note with a branch label (O, I, P, A or B) next to at least two existing questions, for example 'Why do powders behave so differently?' (physical or inorganic), 'Why can some mixtures be separated easily?' (analytical), 'How do medicines work and why is too much dangerous?' (biochemistry and organic). Each group writes one new question that a chemist in a specific branch could investigate, for example 'How do chemists in Nairobi pharmaceutical companies check that every paracetamol tablet contains 500 mg?' (analytical) or 'How can we test which Magadi soda ash is pure?' (analytical or inorganic). Groups explain the label they chose to another group.	Say: 'Our board has many questions. Today we sort them by which branch could help answer them.' Model one example on the board. Give each group 3 sticky notes. Ask: 'Which questions cannot be answered by one branch alone?' WAIT TIME 10-15 seconds. Cold-call 3 groups to read one new question and say which branch would investigate it. Ask the class to respond respectfully: 'Does anyone see a different branch? Tell us why.' Point out that the key inquiry questions (How are medicines for chronic diseases developed? Do all substances have the same properties and uses? Why are food products, detergents and man-made things the way they are?) need several branches. Mark questions that remain unanswered for Lesson 6, where industries and model v2 will be studied.	Question sorting on the DQB (Asking Questions): linking each question to a branch shows learners how a big driving question breaks into smaller investigable questions, and keeps the board as a record of growing understanding.	Observable indicators: (1) every group places at least 2 branch labels on existing questions and writes 1 new question; (2) at least half of the new questions refer to a tray substance (A-H) or mixture M; (3) learners can justify a label in one sentence using evidence. Note questions that are about careers or labels and flag them for Lessons 7 and 8.
Model Building Phase  VIDEO: No closely matching video in the ARES library for this activity. Search ARES for videos Search terms: chemistry branches organic inorganic  EXERCISE: Organic Chemistry Practice Related topic: not an exact match for this lesson. Source: CK-12 Search ARES	Learners open their model notes (class model v1 from Lesson 3 remains the current version) and add a 'Branches' section under the heading 'How chemists study the tray'. Each learner draws or lists a simple diagram with the tray at the centre and five arrows labelled with the five branches, each with one example from the tray (organic: sugar H and paracetamol G; inorganic: salt A and iron E; physical: melting points; analytical: separation of mixture M; biochemistry: action of antacid F and paracetamol G). Learners also write a respect reflection: one idea from another member that improved their group's answer. These notes will feed the revision to model v2 in Lesson 6.	Say: 'Our model v1 explains why substances differ. Now add: who studies which part?' Give 5 minutes for writing. Walk around and ask: 'Which branch would you need to check that two tablets contain the same amount of paracetamol?' WAIT TIME 10-15 seconds. Cold-call 3 learners to read their respect reflection or one diagram label. Close with: 'Which branch do you think helps most with safe use of medicines, and why? Keep your answer for next lesson.' Remind learners that the DQB will be revisited in Lesson 6 when we look at Kenyan industries and revise the model to v2. Collect model notes to check understanding.	Cumulative model building (Developing and Using Models): learners add a new layer (branches) to the existing model notes without discarding earlier ideas, showing that models grow as evidence increases. The respect reflection links the scientific practice of listening to peers' ideas with improving the group's model.	Observable indicators: (1) diagram includes all five branches with one correct tray example each; (2) at least 80% of learners write a specific respect reflection naming an idea from a peer; (3) learners can state in their own words that chemists in different branches study different aspects of the same substances. Use the collected notes to plan reteaching of physical versus analytical in the next lesson opener if needed.

for similar readings				
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D. TEACHER REFLECTION

Did every learner contribute to the group discussion, or did a few dominate? Which branch was hardest to match (for example physical versus analytical), and what evidence helped learners decide? Did learners realise that one substance, such as paracetamol G, can belong to more than one branch? How well did the sentence frames support respectful disagreement? Which DQB questions are still unanswered and need Lesson 6 (importance of chemistry in Kenya and model v2) or later lessons on careers, labels and drugs? What would I change about the station timing or roles next time?

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	We observed data from the Kisumu tray: sugar H and paracetamol G melt and brown at about 186 C and 169 C; salt A dissolves and iron E is pulled out by a magnet; sulphur D melts at about 115 C; mixture M (9 g) was separated by magnet (2.9 g), filtration (2.8 g) and evaporation (2.7 g); the antacid F fizzes in vinegar while paracetamol G does not.
What did I learn?	Chemistry has five branches: organic, inorganic, physical, analytical and biochemistry. Each answers different questions about the same substances, and one substance can belong to more than one branch. Listening respectfully to every group member's opinion improved our answers.
How does this explain the phenomenon?	The tray puzzles are answered by different branches working together: organic (sugar H and paracetamol G), inorganic (salt A and iron E), physical (melting points), analytical (separating mixture M) and biochemistry (how the antacid F and paracetamol G act in the body). This helps chemists make and test medicines, foods and other products safely, which moves us toward answering the driving question.

LESSON 6 (80 minutes (2 x 40-minute periods)): Importance of Chemistry in Kenya: From the Kisumu Tray to Our Industries (Model v2)

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	Learners explore the role of Chemistry in day-to-day life by linking the properties of substances A-H from the Kisumu tray to how chemistry is used in Kenyan agriculture, pharmaceutical, medicine, manufacturing, entertainment, nuclear, food, sports and energy industries, and then revise the class model from v1 to v2.
Knowledge	Learners will know that: (1) industries choose, separate and make materials using properties such as solubility, magnetism, melting point, burning and reaction with acid; (2) the antacid (F) works because it reacts with and neutralises excess stomach acid, while paracetamol (G) relieves pain and fever by a different action and does not neutralise acid; (3) the adult paracetamol maximum is $8 \times 500 \text{ mg} = 4000 \text{ mg} = 4 \text{ g}$ in 24 hours and more can damage the liver; (4) salt (A) and sugar (H) dissolve and cannot be filtered out, which matters in food and sugar industries such as Mumias and Nzoia; (5) sulphur (D) burns with a blue flame and a sharp choking smell (sulphur dioxide), so its use needs care.
Skills	Learners will: develop and use models (revise class model v1 to v2); obtain, evaluate and communicate information (brainstorm and share industry links); construct explanations using tray evidence; analyse and interpret data (calculate $8 \times 500 \text{ mg} = 4000 \text{ mg} = 4 \text{ g}$); collaborate and communicate in groups (KICD core competencies: Communication and Collaboration; Citizenship).
Attitudes	Value: Respect. Learners show acceptance by appreciating the opinion of each group member as they discuss how chemistry is used in different industries. Learners appreciate the importance of chemistry in Kenya and adopt a responsible attitude to medicines and chemicals. PCIs: Life skills (safety and security); Health Promotion Issues (drug and substance use) through safe dosage; Socio-Economic and Environmental issues (consumer protection).
Key Inquiry Question	KICD Key Inquiry Questions: (1) How are medicines for chronic diseases developed? (2) Do all substances around the world have the same properties and uses? (3) Why are food products, detergents and the things in the surrounding especially those that are man-made the way they are? Lesson focus: How does knowing the properties of substances like A-H help Kenyan industries make useful products, and how do we use them safely?
Purpose in Storyline	Lesson 3 gave the class model v1 (classifying A-H as elements, compounds or mixtures) and Lesson 5 taught the five branches of chemistry (organic, inorganic, physical, analytical, biochemistry). Lesson 6 moves from asking what chemistry is to asking why it matters: learners connect tray evidence to real Kenyan industries and revise the model to v2. Lesson 7 will use these industries to explore careers, Lesson 8 will examine product labels, and Lesson 9 will study drugs and substance abuse.
Safety Notes	No tasting of any substance A-H or any product. Goggles on for any demonstration. Sulphur (D) burning is a teacher-only demonstration, in a well-ventilated space or near an open window, with a very small amount in a deflagrating spoon; learners stand back and do not inhale the sharp choking sulphur dioxide fumes. Learners with asthma sit near the door. Tablets F and G are never swallowed in class and are handled as lab materials only. Wash hands after handling. Discuss medicines only as information: never give children adult doses and always follow the label or a health worker's advice.

B. LESSON OVERVIEW

Mrs Achieng returns to the Kisumu tray (A salt, B water, C sand, D sulphur, E iron filings, F antacid tablet, G paracetamol tablet, H sugar) and asks: 'If these eight ordinary things behave so differently, who in Kenya uses these differences, and why?' Learners predict which industries rely on each property, then gather evidence

from short Kenyan industry cards (Kericho tea, Magadi soda ash, Mumias/Nzoia sugar, KenGen Olkaria geothermal, Kenya Pipeline, Nairobi pharmaceutical manufacturers, nyama choma and ugali) and a quick teacher-led sulphur demonstration. They explain how each industry uses properties, work out the safe paracetamol maximum ($8 \times 500 \text{ mg} = 4 \text{ g}$ in 24 hours), add new questions to the Driving Question Board, and revise model v1 to v2 by adding an 'Industries and uses' layer and a 'Safety' layer. This advances the driving question about how properties help chemists make medicines, foods and other products and how to use them safely.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase  VIDEO: <i>No closely matching video in the ARES library for this activity.</i> Search ARES for videos Search terms: chemistry importance agriculture pharmaceutical  HTML: Energy in Chemistry (Flexbook) <i>Related topic: not an exact match for this lesson.</i> Source: CK-12 Search ARES for similar readings	(About 12 minutes) Groups of four (e.g. Group 1: Otieno, Wanjiru, Amina and Kipchirchir) look at their tray A-H and a list of nine industries: agriculture, pharmaceutical, medicine, manufacturing, entertainment, nuclear, food, sports and energy. Individually they write for 2 minutes: 'Which industry do you think uses the property of one tray substance? Choose at least three substances.' Example prompts: salt and sugar dissolving and food; sulphur burning and energy or manufacturing; the antacid reacting with acid and medicine. Groups then agree on one prediction and record it on a Predict card: 'We think ____ industry uses ____ because ____.' They also predict: 'Why might the same chemical be both useful and dangerous?'	Hold up tray A-H and say: 'Last time we met the five branches. Today we ask: who in Kenya depends on the differences we saw at this tray?' Read the nine industries aloud. Say: 'Write silently first. There are no wrong predictions today.' WAIT TIME 10-15 seconds after the question 'Which industry might use the way salt and sugar dissolve?' before taking answers. Cold-call 4 learners from different groups (not volunteers) to share one prediction; ask a different learner to restate each: 'Who can say that in your own words?' Say: 'Respect: listen to every idea in your group before choosing one.' Circulate and note common ideas; do not correct yet. Post two or three predictions on the board under the heading 'Our v1 ideas about use'.	Predict-and-Record with individual silent writing before group talk (Think-Write-Pair-Share). Making private ideas visible lets every learner commit to a prediction, which creates a reason to look for evidence and gives a baseline for noticing how thinking changes. Connecting to Lesson 3's model v1 reminds learners the model so far only classifies A-H.	Observe whether each learner writes at least three substance-industry links independently within the 2 minutes; check Predict cards for a property named (e.g. dissolves, burns, reacts with acid) rather than only an industry name; tally on a checklist how many groups link a property to a use. Look for learners who invite every member's idea (Respect indicator).
Observe Phase  VIDEO: <i>No closely matching video in the ARES library for this activity.</i> Search ARES for videos Search terms: chemistry importance agriculture pharmaceutical	(About 25 minutes) Learners rotate through three evidence stations (8 minutes each) in their groups. STATION 1 - Industry cards: short Kenyan cards on agriculture (fertilisers and soil testing for Kericho tea), manufacturing (Magadi soda ash and detergents), food (Mumias/Nzoia sugar: sugar H dissolves, and salt A as a preservative and flavour for nyama choma and ugali), energy	Before stations say: 'You are chemists collecting evidence. Every claim needs a piece of evidence from the tray or the card.' Set a 8-minute timer per station. At Station 2 ask: 'Which tablet works by reacting with acid? How do you know from the fizz?' WAIT TIME 10-15 seconds before taking responses. Ask: 'If a learner takes 10 tablets of 500 mg in one day, how many mg is that and how	Evidence Table with Claim-Evidence Linking (stations rotation). Learners connect each industry use to a measurable property they have already observed on the tray (dissolving, magnetism, melting, burning, reaction with acid), turning abstract 'importance of chemistry' into specific, testable links. Repeated retrieval of observations from Lessons 2-3 deepens	Check the evidence tables: at least 6 of 9 industries filled with a named substance or property. Listen for correct statements such as 'The antacid fizzes in vinegar because it reacts with acid' and 'Paracetamol did not fizz so it does not neutralise acid'. Check the calculation $8 \times 500 \text{ mg} = 4000 \text{ mg} = 4 \text{ g}$ is correct and units are written. Note who records the sulphur observations (blue flame,

 EXERCISE: Energy in Chemistry Practice <i>Related topic: not an exact match for this lesson.</i> Source: CK-12  Search ARES for similar readings	(KenGen geothermal at Olkaria and Kenya Pipeline fuels), sports (energy drinks and sports-injury creams), entertainment (colours in fireworks and stage lighting), and nuclear chemistry (radioisotopes used in hospitals and in research). Learners fill a table: Industry Substance or property from tray How chemistry helps. STATION 2 - Medicine facts card: antacid (F) neutralises excess stomach acid, relieving heartburn; paracetamol (G) relieves pain and fever and does not neutralise acid; adult label: 1 to 2 tablets of 500 mg every 4 to 6 hours, maximum 8 tablets in 24 hours. Learners calculate $8 \times 500 \text{ mg} = 4000 \text{ mg} = 4 \text{ g}$ and record: 'More than this can damage the liver.' They re-observe F fizzing in dilute vinegar and G giving no fizz (teacher-prepared samples). Learners also read that antacid taken too often can cause constipation or diarrhoea. STATION 3 - Teacher-led demonstration (watched from a safe distance, goggles on): a very small amount of sulphur (D) melts and burns with a blue flame and a sharp choking smell (sulphur dioxide). Learners record what they see and smell from a distance, and recall that sand (C) did not change when heated, whereas sulphur burns. They link this to manufacturing and energy industries that use burning reactions carefully.	does it compare to 4000 mg?' ($10 \times 500 = 5000 \text{ mg}$, which is 1000 mg over the limit). Cold-call 3 groups to share their calculation. For the sulphur demonstration, put on goggles, use a tiny amount in a deflagrating spoon by an open window, and say: 'Stay behind the line. Do not sniff directly. What do you notice about the flame and smell?' Cold-call 2 learners to describe. Remind: 'Never taste. Never give a child an adult dose.' Circulate to make sure every learner records in their own notebook.	understanding of properties as the reason for uses.	sharp choking smell) accurately.
Explain Phase  VIDEO: <i>No closely matching video in the ARES library for this activity.</i>  Search ARES	(About 18 minutes) Each group writes a Claim-Evidence-Reasoning (CER) paragraph answering: 'How does knowing the properties of substances help Kenyan industries make useful products and keep people safe?'	Model one sentence stem on the board: 'Claim: ____ industry uses the property of _____. Evidence: In our tray, _____. Reasoning: This matters because _____. Say: 'Use at least one number or observation from our tray, such as 4 g or the	Claim-Evidence-Reasoning (CER) with Gallery Swap peer feedback. Writing CER forces learners to connect industry uses to observed properties and quantitative data, while peer star-and-wish feedback exposes gaps and builds	Use a 3-point rubric on each CER: (1) names a correct industry use, (2) cites tray evidence (observation or number), (3) explains reasoning using a property. Look for correct statement that F reacts with and

for videos Search terms: chemistry importance agriculture pharmaceutical  HTML: Energy in Chemistry (Flexbook) <i>Related topic: not an exact match for this lesson.</i> Source: CK-12  Search ARES for similar readings	They must use at least three tray substances (e.g. F and G for medicine; A and H for food; D for burning and energy; E and C for separation and manufacturing) and at least two industries. Groups include the idea that the antacid works because it reacts with and neutralises acid, while paracetamol relieves pain and fever by a different action, and that too much of either is harmful. Then each group joins another group in a Gallery Swap: they read the neighbouring CER and give one 'star' (a strong point) and one 'wish' (a missing link), using respectful language. Finally, as a class, learners help Wanjiru answer: 'Why does the antacid relieve heartburn while paracetamol relieves pain?' using the evidence.	blue flame.' WAIT TIME 10-15 seconds after asking: 'What would go wrong if a pharmacist did not know that paracetamol is harmful above 4 g in 24 hours?' Cold-call 4 learners from different groups during the whole-class debrief; ask each: 'What evidence supports that?' Prompt for respect: 'How did your partner's idea change yours?' Ask: 'Salt A and sugar H both dissolve. How does that help a food or sugar factory such as Mumias or Nzoia?' Probe: 'Why can we not filter out dissolved sugar?' (it needs evaporation). Correct any misconception that all medicines work the same way, that more medicine means faster relief, or that 'natural' means safe. Do not re-teach branch definitions; ask learners to name which branch (from Lesson 5) each example belongs to, e.g. biochemistry for how the drug acts in the body.	collaboration and Respect. Linking to the branches from Lesson 5 shows how different parts of chemistry combine in real industries.	neutralises acid and G does not; correct use of 4 g maximum; correct link 'dissolves so cannot be filtered' for A and H. Note respectful, specific peer comments on swap sheets. Identify groups needing support for the next lesson.
Driving Question Board (DQB) Creation  VIDEO: <i>No closely matching video in the ARES library for this activity.</i>  Search ARES for videos Search terms: chemistry importance agriculture pharmaceutical  EXERCISE: Energy in Chemistry Practice <i>Related topic: not an exact match for this lesson.</i>	(About 10 minutes) Learners revisit the class DQB. In their groups they review the existing questions (e.g. Why do powders behave so differently? Why can some mixtures be separated and others not? How do properties help make medicines?). Each group (a) places a green sticky note next to one question they can now partly answer, writing the evidence (e.g. 'Antacid F reacts with acid so it relieves heartburn'), and (b) writes one new question on a yellow note arising from today, such as: 'How are medicines for chronic diseases developed?', 'Do substances in other countries have the same properties and uses?', 'Why are detergents and food products made the way they are?' or 'How	Say: 'Our board is a record of our thinking. Which of our questions have we moved forward today?' Give each group 3 minutes to decide. WAIT TIME 10-15 seconds before asking: 'Which new question do you think a chemist working in a Nairobi pharmaceutical company would ask?' Cold-call 3 groups to read their new question and explain why it matters. Group similar questions together and link them to the KICD key inquiry questions. Say: 'Careers and labels are coming next, so keep these questions in mind.' Ensure every group has placed at least one answered and one new question.	Driving Question Board Tracking. Visibly sorting questions into answered and new shows progress, shows that science is an ongoing process of asking questions, and gives learners ownership of the unit direction. Connecting to KICD inquiry questions keeps the unit coherent.	Check that each group has placed at least one green note supported by tray or industry evidence and one new question. Look for questions that move beyond recall (why/how) and for any link to safety. Note which questions refer to chronic-disease medicines or product design, to plan Lessons 7-9.

Source: CK-12 Search ARES for similar readings	do we know a medicine is safe?'. Groups place notes on the DQB under headings: Answered so far / New questions. The class picks two questions to follow into Lesson 7 (careers) and Lesson 8 (labels).			
Model Building Phase  VIDEO: No closely matching video in the ARES library for this activity. Search ARES for videos Search terms: chemistry importance agriculture pharmaceutical  HTML: Energy in Chemistry (Flexbook) Related topic: not an exact match for this lesson. Source: CK-12 Search ARES for similar readings	(About 15 minutes) Learners look at the class model v1 from Lesson 3 (classification of A-H into elements, compounds and mixtures). In groups they propose what to add so the model also explains why chemistry is important. As a class they build model v2 with two new layers: (1) 'Industries and uses' layer: arrows from properties (dissolves, magnetic, melts, burns, reacts with acid) to industries (food: A and H; agriculture; manufacturing: soda ash from Magadi; energy: Olkaria and Kenya Pipeline; medicine and pharmaceutical: F and G; sports; entertainment; nuclear), and (2) 'Safety' layer: arrows showing labels, correct dosage (maximum 4 g paracetamol in 24 hours), avoiding substance misuse and careful handling of sulphur burning. Each group draws a mini-version in their notebook and writes a 2-sentence caption answering the driving question: 'How do the different properties of substances help chemists make medicines, foods and other products, and how do we use them safely?' Groups mark what in v2 is still uncertain (e.g. how medicines for chronic diseases are developed).	Display v1 and say: 'v1 told us what the substances are made of. v2 must tell us what they are used for and how to stay safe. What should we add?' WAIT TIME 10-15 seconds before taking suggestions. Cold-call 4 learners to propose one addition each and ask the class: 'Do we agree? What evidence supports it?' Draw the new layers with learner input, requiring each arrow to be justified by evidence from the tray or the industry cards. Ask: 'Which part of our model explains why the antacid and paracetamol are different?' and 'Where does safety belong?' Label the model 'v2 (Lesson 6)' and say: 'We will revise it again to reach the final model in Lesson 10.' Collect captions for review.	Cumulative Model Revision (v1 to v2). Adding layers to an existing model shows how new evidence extends earlier ideas instead of replacing them, and justifying every arrow with evidence builds the habit of linking claims to data. Marking uncertainties creates a bridge to later lessons.	Check each mini-model for: (1) properties linked to at least four industries, (2) a safety layer that includes the 4 g paracetamol limit and avoiding misuse, (3) an evidence-based caption answering the driving question. Look for learners who can explain at least one arrow to a peer. Identify any misconception (e.g. that antacid and paracetamol do the same job) for follow-up in Lesson 8 and 9.

D. TEACHER REFLECTION

After the lesson, reflect: (1) Could learners link at least three tray substances to specific industries using a property as evidence? (2) Did every learner correctly calculate

8 x 500 mg = 4000 mg = 4 g and explain why exceeding it is dangerous? (3) Did learners explain that the antacid (F) reacts with and neutralises acid while paracetamol (G) acts on pain and fever differently? (4) Which industry examples were unfamiliar, and how can I add local context (e.g. a nearby factory or farm in Kisumu) next time? (5) Did all voices contribute respectfully, and who did I not hear from? (6) How well did model v2 capture both uses and safety, and what do learners still wonder about for careers (Lesson 7) and labels (Lesson 8)? Note any safety issues during the sulphur demonstration and plan improvements.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	Record what you observed: the properties of A-H (salt and sugar dissolve; sand and iron do not; iron is pulled by a magnet; sulphur melts and burns with a blue flame and sharp choking smell; F fizzes in dilute vinegar but G does not), and the Kenyan industry examples you studied.
What did I learn?	Record what you learned: chemistry is important in agriculture, pharmaceutical, medicine, manufacturing, entertainment, nuclear, food, sports and energy industries because industries choose, separate and make materials using properties; the antacid neutralises acid, paracetamol relieves pain and fever, and the adult maximum is 8 x 500 mg = 4 g in 24 hours.
How does this explain the phenomenon?	Explain how this helps answer the driving question: use evidence from the tray and at least two industries to explain how properties help chemists make medicines, foods and other products, and how labels, correct dosage and avoiding substance misuse keep people safe. Describe what you added to model v2.

LESSON 7 (40 minutes (one double lesson of 80 minutes is recommended if digital search is used)): Careers in Chemistry: Who Works with the Kisumu Tray Substances, and Who Gets to Choose?

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	Learners search print and digital media for careers related to Chemistry, link each career to the Kisumu tray evidence (antacid F, paracetamol G and the separation of mixture M), examine how gender stereotypes influence career choices, and present findings in plenary.
Knowledge	Chemistry careers (for example pharmacist, analytical chemist, quality control officer, chemical engineer, food technologist, laboratory technician, drug inspector, geothermal chemist) each use the properties of substances (solubility, magnetism, melting point, reaction with acid) that learners saw in the tray. A gender stereotype is a fixed idea about what girls or boys can or should do; it can limit career choice but does not change who can do chemistry.
Skills	Searching for information in print and electronic media; evaluating sources; collaborating in groups; presenting findings in plenary; constructing evidence-based explanations (NGSS SEP: Obtaining, Evaluating and Communicating Information; Engaging in Argument from Evidence).
Attitudes	Respect (KICD value): the learner shows acceptance while appreciating the opinion of each member in group discussion. Self-efficacy: confidence in presenting findings. Citizenship: critical and constructive dialogue. Openness to careers in chemistry regardless of gender.
Key Inquiry Question	How are medicines for chronic diseases developed? Who are the people that do this work, and what properties of substances do they use? (KICD key inquiry 1, with a link to key inquiry 3 on how products are made.)
Purpose in Storyline	Lessons 5 and 6 gave us the branches of chemistry and the industries that use chemistry in Kenya. Lesson 7 asks who works in those branches and industries and what they do with the properties we found in the tray. It advances the driving question by showing that people with different careers use properties to make medicines, separate mixtures and keep products safe. It also prepares Lesson 8 (quality control and labels) and Lesson 9 (dosage and drugs).
Safety Notes	No new heating or chemical tests. Trays A-H may be displayed sealed and must not be tasted; tablets F and G stay in their packets as props. Learners with devices use only school-approved or offline resources; the teacher supervises searches. Treat all learner opinions respectfully; no ridiculing of any learner's career dreams. Hands washed if trays were handled.

B. LESSON OVERVIEW

Mrs Achieng brings the Group 1 tray (A salt, B water, C sand, D sulphur, E iron filings, F antacid tablet, G paracetamol tablet, H sugar) and mixture M back to the front. She asks: 'Someone had to decide that F relieves heartburn and G does not, and someone had to separate and check what is in a product. Who are these people?' Learners predict careers, then research in groups from print (career booklets, newspapers, textbooks) and offline or digital resources, matching each career to a tray property. Next they examine a short set of career profiles and survey data to discuss how gender stereotypes shape choices, and each group presents in plenary. The class adds new questions to the DQB and records how careers help answer the driving question, in readiness for revising the model in Lesson 10.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase  VIDEO: No closely matching video in the ARES library for this activity. Search ARES for videos Search terms: chemistry careers pharmacist analytical  READING: No closely matching reading in the ARES library for this activity. Search ARES for readings Search terms: chemistry careers pharmacist analytical	Learners look at the tray (A-H) and mixture M displayed in Group 1's setup. In their notebooks they individually write: (1) three jobs in which a person might use the properties of F antacid or G paracetamol or the separation of mixture M (magnet, filtration, evaporation); (2) one prediction: 'Do you think more men or more women work as chemists in Kenya? Why?' Then they share in groups of four (like Otieno, Wanjiru, Amina and Kipchirchir) and agree on a group list of five possible careers.	Hold up tablet packets F and G and say: 'Wanjiru asked why the antacid relieves heartburn but paracetamol relieves pain. Who in Kisumu or Nairobi might answer that question for a living?' Give 1 minute silent writing, then WAIT TIME of 10-15 seconds before any hands go up. Say: 'Turn to your group; every voice must be heard, and we thank each person for their idea.' Cold-call 4 groups (not volunteers) to give one career each, and record on the board without correcting. Ask one neutral question: 'What made you think of that job?' Do not comment on the gender prediction yet; just record the class tally on the board.	Think-Pair-Share with a recorded prediction. Writing alone first activates prior knowledge from Lessons 5 and 6 (branches and industries) and exposes learners' initial ideas, including stereotypes, so that they can be compared with evidence later.	Look for: each learner has written three jobs and one reasoned prediction; at least 6 of 8 groups name a career linked to medicine or testing (for example pharmacist, laboratory technician); groups give reasons that mention a property or an industry from Lesson 6. Note learners who give gendered reasons (for example 'chemistry is for boys') for use in the Explain phase, without naming them.
Observe Phase  VIDEO: No closely matching video in the ARES library for this activity. Search ARES for videos Search terms: chemistry careers pharmacist analytical  READING: No closely matching reading in the ARES library for this activity. Search ARES for readings Search terms: chemistry careers pharmacist analytical	Groups receive a Career Research Card and spend about 15 minutes gathering evidence from print (career booklets, textbooks, newspaper cut-outs, KCSE career guides) and, where available, school-approved digital sources or offline resource files. Each group researches three careers: (a) pharmacist or pharmaceutical scientist (relates to F and G and the Pharmacy and Poisons Board), (b) analytical chemist or laboratory technician (relates to separating mixture M: magnet, filtration, evaporation, with 2.9 g, 2.8 g and 2.7 g recovered and 0.6 g lost out of 9 g), (c) one free choice such as quality control officer at KEBS, food technologist (Mumias/Nzoia sugar, Kericho tea), chemical engineer (Magadi soda ash), geothermal chemist (KenGen,	Circulate with the question: 'Which property from our tray does this person use, and how do you know?' Prompt: 'Show me the line in your source that says so.' For groups with only digital sources, ask: 'Who wrote this and can we trust it? What is a second source?' Stop the whole class at 8 minutes: 'Pens down. Recorders, read one row aloud.' Use 10-15 seconds WAIT TIME after asking: 'Is the job described only for men or only for women, or does the source just say a person?' Cold-call 3 groups for one piece of evidence each. Remind: 'Appreciate each person's contribution; use the phrase "I agree with... because..." or "I see it differently because..."'.	Evidence table with source evaluation (Obtaining, Evaluating and Communicating Information). Linking each career to a tray property or separation step makes the abstract job concrete and shows that careers apply the same properties that explain the phenomenon.	Check the tables: each group has three careers with a named property or separation step (for example 'filtration removes sand C; analytical chemist checks purity') and a branch; at least one group notes that the filtrate evaporation recovers salt A (2.7 g). Observe collaboration: all four learners contribute to the table, roles rotated, and respectful phrases are used. Collect any digital-literacy problems such as accepting the first search result without checking.

	Olkaria). For each, they fill a table: career, what the person does, which tray property or separation method is used, branch of chemistry from Lesson 5, and one example of a Kenyan employer. Roles in the group: reader, recorder, searcher, presenter, rotated. Second, groups read three short career profiles (for example a woman analytical chemist in a Nairobi pharmaceutical company and a man food technologist) supplied by the teacher as examples, and note any statement that shows a stereotype.			
Explain Phase  VIDEO: No closely matching video in the ARES library for this activity. Search ARES for videos Search terms: chemistry careers pharmacist analytical  READING: No closely matching reading in the ARES library for this activity. Search ARES for readings Search terms: chemistry careers pharmacist analytical	Learners use their evidence to explain two ideas in groups, then in class. First: 'How does a career in pharmaceutical or analytical chemistry help us answer why F relieves heartburn and G relieves pain, and why too much of either is harmful?' They write a claim-evidence-reasoning paragraph using MEDICINE FACTS: antacid neutralises excess stomach acid; paracetamol relieves pain and fever and does not neutralise acid; adult paracetamol maximum is 8 tablets of 500 mg = 4000 mg = 4 g in 24 hours; more can damage the liver. Second: they compare their Predict-phase gender tally with the career profiles and discuss 'What is a gender stereotype? How might it stop a girl or a boy from choosing chemistry? What does the evidence show about who can do these jobs?' Groups then prepare a 2-minute plenary presentation with one poster sheet and present, with one group member from each group speaking and a different member answering questions.	Model a sentence frame on the board: 'A [career] uses [property] to [action], which helps [people or product].' Ask: 'Why is it not enough for a pharmacist to know that F fizzes in vinegar?' WAIT TIME 10-15 seconds. Then: 'What do labels and dosage tell us, and who writes and checks them?' (Preview of Lesson 8, not taught here.) For stereotypes, define: 'A gender stereotype is a fixed idea about what girls or boys should do.' Ask: 'Where do we see these ideas in Kisumu, in the media, in families?' WAIT TIME 10-15 seconds, then cold-call 4 learners from different groups. Say: 'We disagree with ideas, never with people.' During plenary, cold-call 2 listeners per presentation to restate one point and ask: 'What evidence supports that?' Do not allow stereotypes to be stated as facts; redirect: 'What evidence do we have for that?' Ensure girls and boys both speak.	Claim-Evidence-Reasoning plus Structured Academic Controversy-style discussion (Engaging in Argument from Evidence). CER ties careers to the tray data and medicine facts; the stereotype discussion has learners test a claim against evidence from profiles and their own tally, building critical and constructive dialogue and self-efficacy.	Listen for CER paragraphs that include a claim, at least two pieces of evidence (for example 'F fizzes in vinegar, so it reacts with acid' and '8 x 500 mg = 4 g maximum') and reasoning linking to a career. In the stereotype discussion, look for learners who define a stereotype in their own words, give a reason that is evidence-based, and show respect by building on another's idea. In plenary, check that presenters speak clearly and that at least 6 of 8 groups present accurate property-career links. Note any learner still holding 'chemistry is for boys/girls' and plan a follow-up conversation.

Driving Question Board (DQB) Creation  VIDEO: No closely matching video in the ARES library for this activity. Search ARES for videos Search terms: chemistry careers pharmacist analytical  READING: No closely matching reading in the ARES library for this activity. Search ARES for readings Search terms: chemistry careers pharmacist analytical	Learners return to the class DQB. Each group writes on sticky notes (1) one thing the careers research helped answer about the driving question, placed under 'We have evidence for...', and (2) one new question, for example 'How long does it take to develop a medicine?' or 'What subjects do I need to become a pharmacist in Kenya?' They place notes on the DQB, moving or linking earlier questions such as Wanjiru's about F and G.	Say: 'Look at the board. Which earlier question can a career now help answer?' WAIT TIME 10-15 seconds. Cold-call 3 learners to move or connect a question and explain why. Ask: 'Which of our questions needs a person with a different job, such as a drug inspector or quality control officer?' Mark questions about labels, quality marks and drug dosage with a star: 'We will return to these in Lessons 8 and 9.' Keep note of questions for the model revision.	Gallery sort of the DQB. Sorting questions into answered, partly answered and new makes learning visible, keeps the phenomenon at the centre and tells learners where the storyline goes next.	Check the sticky notes: each group posted one evidence note and one new question; at least half the new questions link to a career or to product safety; learners can justify where they placed a note. Observe whether learners connect Wanjiru's question to pharmaceutical and analytical roles.
Model Building Phase  VIDEO: No closely matching video in the ARES library for this activity. Search ARES for videos Search terms: chemistry careers pharmacist analytical  READING: No closely matching reading in the ARES library for this activity. Search ARES for readings Search terms:	Learners open their personal copy of the class model v2 from Lesson 6 and add a section titled 'People who use properties'. With a drawing or flow chart they show how a pharmacist or pharmaceutical scientist, an analytical chemist or technician and a quality control officer connect the properties of A-H (solubility, magnetism, melting point, reaction with acid) to making, separating and checking products such as F antacid and G paracetamol. They label where safety (dosage, label, avoiding substance misuse) fits. The model remains a v2 draft with additions marked in a different colour; it will be finalised in Lesson 10.	Say: 'Add a new part to your model: the people. Arrows should show property, then job, then product or safety.' Give 10-15 seconds WAIT TIME before groups begin drawing. Circulate and ask: 'Where does the 0.6 g loss in mixture M matter for an analytical chemist?' and 'Where in your model does a person keep someone safe from too much paracetamol?' Cold-call 3 groups to explain one arrow. Ask: 'Does your model show that a career is open to everyone, whatever their gender?' Collect models for the next lesson.	Model revision with added components. Adding people and decisions to the property-based model shows that chemistry knowledge is applied by humans, which helps learners build towards a full answer to the driving question and a view of themselves in chemistry.	Look for models with at least three labelled arrows linking a property or separation step to a career and a product or safety outcome (for example 'magnet removes iron E: lab technician checks purity'); correct use of tray letters and names; revisions marked in a new colour. Exit ticket: 'Name one career and the tray property it uses.' Success: at least 80% of learners name a matched pair.

chemistry careers pharmacist analytical				
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D. TEACHER REFLECTION

Which learners offered stereotypes, and were they challenged with evidence rather than embarrassed? Did every group find a clear link between a career and a property from the tray (F, G or mixture M)? Were all four group members, including quieter learners, speaking during research and presentation? Which sources did learners trust without checking, and how will I teach source evaluation next time? Which DQB questions about labels and dosage should open Lesson 8? Which learners need support to see themselves in chemistry careers?

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)	
What did I observe?	We observed that careers such as pharmacist, analytical chemist, laboratory technician and quality control officer use the properties of substances we saw in the tray: F antacid reacts with acid, G paracetamol does not, and mixture M was separated by magnet, filtration and evaporation (2.9 g, 2.8 g and 2.7 g recovered out of 9 g).
What did I learn?	We learned that chemistry offers many careers in medicine, industry and testing, and that gender stereotypes are fixed ideas that can wrongly limit who chooses these careers.
How does this explain the phenomenon?	We can explain that people in chemistry careers use properties to make, separate and check products such as medicines, and that safety through labels and correct dosage matters, and that girls and boys can both choose these careers; this adds to our answer to the driving question.

LESSON 8 (40 minutes): Consumer Protection and Product Labels: Reading the Story a Label Tells

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	Learners read and interpret labels on manufactured products (including the paracetamol 500 mg label from the Kisumu tray) to identify quality marks, expiry dates, ingredients and preservatives, nutritional information and caution messages, and explain how this information protects consumers.
Knowledge	A label is the chemist's and manufacturer's message to the consumer. The KEBS Diamond Mark shows a product has met quality standards. The expiry date shows when a product may no longer be safe or effective. Ingredients and preservatives show what a product is made of. Caution messages and dosage instructions prevent harm. The paracetamol label (adult: 1 to 2 tablets of 500 mg every 4 to 6 hours, maximum 8 tablets = 4000 mg = 4 g in 24 hours) uses the properties of substance G to keep people safe, because more than this can damage the liver. Antacid F also carries a label: taking it too often can upset the body's salts and cause constipation or diarrhoea. Children must not be given adult doses.
Skills	Observing and recording information from labels; asking questions (SEP 1); obtaining, evaluating and communicating information (SEP 8); using mathematics to check dosage ($8 \times 500 \text{ mg} = 4000 \text{ mg} = 4 \text{ g}$) (SEP 5); constructing explanations (SEP 6); developing and using models (SEP 2).
Attitudes	Respect: learners show acceptance while appreciating the opinion of each member as they discuss label information in groups. Learners develop a habit of reading labels before buying or using any product. Citizenship: learners recognise the consumer's right to information on drug prescription, substance use and side effects.
Key Inquiry Question	Why are food products, detergents and the things in the surrounding, especially those that are man-made, the way they are, and how does the information on their labels help consumers use them safely?
Purpose in Storyline	Lesson 6 showed how chemistry is used in industries and products, and Lesson 7 introduced careers in chemistry. This lesson moves from who makes products to how consumers are protected, by reading labels. Learners see that the properties studied in Lessons 1 to 3 (for example, paracetamol G and antacid F) are communicated through labels, and that quality control chemists check them. It prepares Lesson 9 (drugs, prescription, dosage and substance abuse), where dosage information is explored in depth.
Safety Notes	No tasting or smelling of any product or tablet. Tablets F and G remain sealed or are handled only with clean hands and are NOT swallowed. Use empty, clean, washed containers for food and detergent samples. Do not open detergents or chemical products. Wash hands after the activity. Do not give anyone medicines in class. Wear goggles only if any demonstration with the tray is repeated. Explain that expired medicines are returned to a pharmacy or health facility, not thrown away or shared.

B. LESSON OVERVIEW

Otieno comes to class from his family's kiosk with a worrying story: a customer asked for paracetamol (like tablet G on Mrs Achieng's Kisumu tray) and wanted to give an adult dose to a small child. Learners predict what a label should tell a buyer, then examine sample containers (a paracetamol 500 mg box, an antacid pack, a packet of maize flour, a juice carton and a detergent bottle, all empty or sealed) for the KEBS Diamond Mark, expiry date, ingredients and preservatives, nutritional information and cautions. They explain how each label item protects consumers, check the maximum paracetamol dose ($8 \times 500 \text{ mg} = 4 \text{ g}$ in 24 hours), add new questions to the Driving Question Board, and revisit the class model (v2) to note where labels fit in. The lesson advances the driving question by showing how knowledge of properties is turned into safe use.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase  VIDEO: <i>No closely matching video in the ARES library for this activity.</i> Search ARES for videos Search terms: chemistry label quality mark  READING: <i>No closely matching reading in the ARES library for this activity.</i> Search ARES for readings Search terms: chemistry label quality mark	Learners listen to Otieno's kiosk scenario: a customer wants to buy paracetamol and give an adult dose to a small child, and another customer asks if a packet of old biscuits is still fine. In Group 1 (Otieno, Wanjiru, Amina and Kipchirchir) and the other 7 groups, learners individually write in their notebooks: (1) three things they think a label on a medicine or food must tell the buyer, and (2) a prediction: 'If a customer reads only the label on tablet G (paracetamol 500 mg), what will they know about how to use it safely?' Groups then share and combine their predictions on a group chart under the headings 'Must be on a label' and 'Why it matters'. Learners link back to the Kisumu tray: G and F look like ordinary tablets, so how would a person know which one to use and how many?	Greet learners and say: 'Today we leave the tray for a moment and look at what the shopkeeper sells. Otieno, tell us about your kiosk.' Invite Otieno to share the story briefly (2 minutes). Ask: 'Tablet G and tablet F look like simple white tablets. How would a buyer know what each one is for and how much to take?' WAIT TIME 10-15 seconds before taking any answer. Give 3 minutes of silent individual writing, then 4 minutes of group sharing. Cold-call 4 learners (one from each of 4 different groups) to read one 'must be on a label' item, and write all items on the board without correcting yet. Ask: 'What would happen if a label had no expiry date?' Cold-call 2 more learners. Circulate and note which groups mention quality marks, dosage and cautions, and which mention only the product name.	Eliciting prior knowledge through individual-then-group prediction (Think-Write-Share). Making predictions visible on the board builds understanding because learners commit to ideas that they will test against real labels in the Observe Phase, which creates a need to know and a clear comparison point for later revision.	Observable indicators: (1) At least 6 of 8 groups list three or more label items on their chart. (2) Count how many learners mention dosage or caution (target: at least half of the class). (3) Note learners who mention expiry date or the KEBS Diamond Mark without prompting. (4) Individual notebook check: each learner has written a prediction about tablet G with a reason. Learners with only the product name need a prompt: 'What would a mother need to know to give this to her child safely?'
Observe Phase  VIDEO: <i>No closely matching video in the ARES library for this activity.</i> Search ARES for videos Search terms: chemistry label quality mark  READING: <i>No closely matching reading in the ARES library for this activity.</i>	Each of the 8 groups receives a label kit of clean, empty or sealed containers: a paracetamol 500 mg box (tablet G), an antacid pack (tablet F), a maize flour packet (ugali), a bottled juice or milk carton, and a detergent bottle. A label observation table has these columns: Product KEBS Diamond Mark? Expiry date Ingredients / preservatives Nutritional information Caution or dose message. Learners take roles (reader, recorder, checker, reporter), rotate the roles after 2 products, and use a hand lens or magnifying glass where needed. They record exactly what is	Distribute kits and say: 'You are consumer protection inspectors from the Kenya Bureau of Standards. Read exactly what is printed. Do not taste, smell or open anything.' Show the KEBS Diamond Mark on an enlarged poster or sample. Remind learners of the rule: 'No tasting, no swallowing tablets, wash hands afterwards.' Circulate for 15 minutes and ask: 'Where on the pack did you find the expiry date? What does the date tell you that your eyes cannot?' WAIT TIME 10-15 seconds. Ask groups: 'What does the caution on the paracetamol box tell you about the	Structured data collection with a label observation table (Obtaining information, SEP 8). Using the same table headings for every product allows patterns to emerge across foods, detergents and medicines, so learners can see that different product types carry different information because of different risks.	Observable indicators: (1) Each group's table has at least 4 products with at least 4 of the 6 columns completed accurately. (2) Learners correctly copy the paracetamol dose (1 to 2 tablets of 500 mg every 4 to 6 hours, maximum 8 tablets in 24 hours). (3) Learners distinguish preservatives from active ingredients when asked. (4) At least 6 groups record at least one case of missing or unclear information. (5) Roles were rotated and every learner recorded or read at least once; note any learner not participating.

 Search ARES for readings Search terms: chemistry label quality mark	written, for example the paracetamol label: adult dose 1 to 2 tablets of 500 mg every 4 to 6 hours, maximum 8 tablets in 24 hours, and a warning that more can damage the liver. They note that the antacid label warns about taking it too often (can cause constipation or diarrhoea) and advises following the label or a health worker's advice. They compare food labels (maize flour, juice) which list ingredients, preservatives and nutritional information with the medicine labels, which list active ingredient, dose and cautions. Learners also spot where a product has NO mark or date and record this as 'missing information'.	property of substance G in the body?' Cold-call 3 recorders to report one finding they found surprising. Prompt rotation of roles at the 7 minute mark. For groups that finish early: 'Find a product with a preservative and tell me what you think it prevents.' For learners who need support, give a labelled diagram of a sample label with arrows. Collect observations on a class table on the board.		
Explain Phase  VIDEO: <i>No closely matching video in the ARES library for this activity.</i>  Search ARES for videos Search terms: chemistry label quality mark  READING: <i>No closely matching reading in the ARES library for this activity.</i>  Search ARES for readings Search terms: chemistry label quality mark	Groups use their label tables to explain how each label item protects consumers. Individually, then in groups, learners complete a Claim-Evidence-Reasoning (CER) statement: 'Claim: Labels protect consumers. Evidence: (from our table). Reasoning: (why this item matters).' Each group chooses one item (quality mark, expiry date, ingredients and preservatives, nutritional information, or cautions) and explains it using the paracetamol label as a worked case. They calculate the daily maximum: 8 tablets x 500 mg = 4000 mg = 4 g, and answer Otieno's kiosk question: 'A customer wants to give a small child 2 tablets of 500 mg. Why is this unsafe even though it is within the adult dose?' (Children must not be given adult doses.) They also explain why antacid F and paracetamol G need different labels (antacid neutralises excess stomach acid to relieve	Say: 'Now we move from reading to explaining. A label is evidence. What does your evidence say?' Model one CER on the board with the expiry date. Give 6 minutes for group writing. Circulate and ask: 'How do you know that 4 g is the limit? Show me the calculation.' WAIT TIME 10-15 seconds. Probe: 'If a label says maximum 8 tablets in 24 hours, what do you think happens if someone takes 12?' (can damage the liver). Ask: 'Why does the label say more than the dose can damage the liver, and not just it is wasteful?' Cold-call 4 groups (one reporter each) to present their CER in 1 minute. After each, ask the class: 'Who can add to or challenge this claim using the evidence on your own table?' Cold-call 3 more learners. Reinforce respect by saying: 'Start your response with: I agree with ... because ...'. Clarify that the KEBS Diamond Mark shows the product met a standard and the Pharmacy	Claim-Evidence-Reasoning (CER) with peer critique (SEP 6, Constructing Explanations, and SEP 7, Engaging in argument from evidence). CER builds understanding because learners must tie each statement to their own label data and explain why the evidence supports it, moving from copying text to understanding purpose. Peer critique makes reasoning visible and corrects errors.	Observable indicators: (1) Each group CER contains a claim, at least one specific piece of evidence from the table, and a reason linking to consumer safety. (2) Learners correctly calculate $8 \times 500 \text{ mg} = 4000 \text{ mg} = 4 \text{ g}$. (3) At least 5 of 8 groups state that adult doses should not be given to children. (4) Learners can state that antacid F neutralises acid and paracetamol G relieves pain and fever, and that both are harmful in excess. (5) Learners use respectful phrases when responding to peers (count the instances). Misconception check: a learner who says that a product without an expiry date is safe forever needs more guidance.

	heartburn; paracetamol relieves pain and fever) and what could go wrong if the two are confused. Learners then discuss in groups how a quality control chemist (Lesson 7) would check that a batch of tablets in a Nairobi factory matches its label. Respect is practised: each learner must acknowledge one idea from a groupmate before adding their own.	and Poisons Board regulates medicines in Kenya. Do not teach dosage and drug abuse in depth, as these belong to Lesson 9: say 'We will study this more deeply in the next lesson.'		
Driving Question Board (DQB) Creation  VIDEO: No closely matching video in the ARES library for this activity. Search ARES for videos Search terms: chemistry label quality mark  READING: No closely matching reading in the ARES library for this activity. Search ARES for readings Search terms: chemistry label quality mark	Learners return to the class Driving Question Board with the driving question: 'How do the different properties of substances around us help chemists make medicines, foods and other products, and how do we use them safely?' Each group writes one 'We now know...' sticky note from today's label evidence (for example: 'The label on G tells us the maximum is 4 g in 24 hours because more can damage the liver') and one new 'We still wonder...' question (for example: 'How do chemists test that the tablets contain exactly 500 mg?' or 'Why do some foods need preservatives and others do not?'). They place notes next to the KICD key inquiry question: 'Why are food products, detergents and the things in the surrounding, especially those that are man-made, the way they are?' The class also sorts questions into those that can be answered in Lesson 9 (drugs, prescription, dosage, substance abuse) and Lesson 10 (safe and healthy learning environment, final model). Wanjiru's earlier question about why the antacid and paracetamol differ is revisited and marked with what the labels add.	Say: 'Our board is growing. Which of our earlier questions did the labels help to answer?' Point to Wanjiru's question about why the antacid relieves heartburn while paracetamol relieves pain. WAIT TIME 10-15 seconds. Give groups 3 minutes to write one 'now know' and one 'still wonder' sticky note. Cold-call 4 groups to read their notes aloud and place them on the board. Ask: 'Which of these new questions will Lesson 9 help us answer?' Cold-call 2 learners to sort the questions. Mark any answered questions with a tick and the evidence source (for example, 'Label table, Lesson 8'). Praise questions that ask how or why, and rephrase yes/no questions.	Driving Question Board as a tool for tracking growing understanding and ownership of learning. Linking each note to evidence and to the next lesson helps learners see learning as a connected storyline, and learner-generated questions keep curiosity and purpose high.	Observable indicators: (1) All 8 groups post one 'now know' statement backed by label evidence. (2) At least 6 groups post a new, testable or researchable question. (3) Learners correctly link at least one note to the driving question or a KICD key inquiry question. (4) Check that 'now know' statements do not claim more than the evidence shows (for example, 'all medicines are safe if the label is present' is incorrect).

Model Building Phase  VIDEO: No closely matching video in the ARES library for this activity. Search ARES for videos Search terms: chemistry label quality mark  READING: No closely matching reading in the ARES library for this activity. Search ARES for readings Search terms: chemistry label quality mark	Learners revisit the class model, currently at version 2 (from Lesson 6), which links properties of substances A-H to their uses in industries. In groups, they add a new layer, 'Labels and consumer protection', as a section on the model: arrows from properties (for example, solubility, burning, reaction with acid) to product uses (antacid F, paracetamol G, food products like sugar H and salt A) to label information (quality mark, expiry date, ingredients and preservatives, dose, caution) to safe use by the consumer. They add Otieno's kiosk as a point where the consumer meets the product, and a quality control chemist as the person who checks that products match their labels. Each group draws or adds the revision on a model chart and records a short caption: 'One thing we added to the model today and why.' Learners also annotate where they are still unsure, to be revisited in Lesson 9 and in the final model in Lesson 10.	Say: 'Our model must now answer a new part of the driving question: how do we use these products safely? Where do labels fit?' Display model v2. Give 8 minutes. Circulate and ask: 'How does the label connect the property of substance G to a safe amount for a person?' WAIT TIME 10-15 seconds. Ask: 'What is missing from our model if a customer cannot read the label?' Cold-call 3 groups to explain one new arrow or box. Ask 2 other learners to say whether the addition makes sense using evidence. Do not yet create the final model; say: 'This is a working revision. In Lesson 9 we add drugs and misuse, and in Lesson 10 we finalise.' Close with the exit question: 'Name one label item you will check the next time you buy a product, and say why.' Take answers from 4 learners by cold-call.	Developing and revising models (SEP 2). Adding a new layer to an existing model builds understanding by showing how new evidence extends rather than replaces earlier ideas, and by connecting the science (properties) to real-world decisions (labels and safe use).	Observable indicators: (1) Each group's model includes at least three label items correctly connected to consumer safety. (2) The model links a property or substance from the tray (for example, G or F) to a label message. (3) At least 6 groups add the role of the quality control chemist or KEBS. (4) Exit responses name a specific label item and a valid reason (for example, 'I will check the expiry date because old medicine may not work or may be unsafe'). (5) Learners identify one remaining uncertainty for Lesson 9.
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D. TEACHER REFLECTION

After the lesson, consider: (1) Did learners read labels carefully, or only copy the product names? Which groups needed prompting to find the expiry date or caution? (2) Did learners correctly calculate $8 \times 500 \text{ mg} = 4 \text{ g}$ and explain why it is the maximum, or did they just repeat the text? (3) How well did learners show Respect in group discussion, for example, acknowledging a groupmate's idea before adding their own? (4) Were any learners unable to read the labels because of language or print size, and how can you support them in Lesson 9? (5) Did the CER statements show a link between the properties of substances F and G and the safety information? (6) Which DQB questions will you carry into Lesson 9, and which learners should lead them? Keep a note of any learners who said that a medicine is safe because it is in a shop, and revisit this idea in Lesson 9.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	We observed the labels on a paracetamol 500 mg box (G), an antacid pack (F), a maize flour packet, a juice carton and a detergent bottle. We found the KEBS Diamond Mark, expiry dates, ingredients and preservatives, nutritional information, and caution and dose messages, and we noticed where information was missing.
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What did I learn?	We learned that labels protect consumers by giving information on quality, how long a product is safe to use, what it contains, and how to use it safely. The paracetamol label says adults take 1 to 2 tablets of 500 mg every 4 to 6 hours, with a maximum of 8 tablets (4000 mg = 4 g) in 24 hours, because more can damage the liver. Antacid taken too often can cause constipation or diarrhoea, and children must not be given adult doses.
How does this explain the phenomenon?	Substances like the antacid (F) and paracetamol (G) have different properties and uses, so their labels give different instructions. Labels turn what chemists know about the properties of substances into safe-use messages for the consumer. This helps answer the driving question about how we use products safely, and prepares us to study drugs, dosage and substance abuse in Lesson 9.

LESSON 9 (40 minutes (1 lesson)): Drugs and Substance Abuse: Why the Dose Makes the Difference

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	Learners use evidence from the Kisumu tray (antacid F and paracetamol G) to explain drug, prescription, dosage and substance use, and to examine the effects of drug and substance abuse in day-to-day life. They then design posters to sensitise peers and the community, advancing the driving question on how properties help chemists make medicines and how we use them safely.
Knowledge	By the end of the lesson, learners should be able to: (a) define drug, prescription, dosage and substance abuse using the class conventions; (b) explain that antacid F neutralises excess stomach acid while paracetamol G relieves pain and fever by a different action, and that both are harmful in excess; (c) calculate the paracetamol maximum of $8 \times 500 \text{ mg} = 4000 \text{ mg} = 4 \text{ g}$ in 24 hours; (d) describe the effects of antacid overuse (upset body salts, constipation or diarrhoea) and of paracetamol overdose (liver damage).
Skills	Making predictions; interpreting label data and doing simple calculations (mg to g); constructing explanations from evidence (NGSS SEP: Constructing Explanations); obtaining, evaluating and communicating information through a poster (NGSS SEP); collaborating in groups of four.
Attitudes	Respect: learners accept and value each member's ideas during group discussion (KICD value). Responsibility: learners advocate for safe medicine use and a healthy learning environment. Citizenship: learners recognise the consumer right to information on drug prescription, substance use and side effects.
Key Inquiry Question	How are medicines for chronic diseases developed? Do all substances around the world have the same properties and uses? Why are food products, detergents and man-made things the way they are? (Lesson focus: How does what we know about the properties of a medicine help us decide a safe dose and avoid harm?)
Purpose in Storyline	Lesson 8 taught how to read labels (quality mark, expiry date, ingredients, cautions) using the paracetamol 500 mg label and dose. Lesson 2 gave the evidence that substances have their own properties, including F reacting with acid and G not reacting. Lesson 9 uses that knowledge to answer Wanjiru's puzzle about why the antacid and paracetamol differ and why too much of either is dangerous. It turns the dosage information into community action through posters, preparing for Lesson 10, where learners finalise the model and consider rights and responsibilities for a safe school environment.
Safety Notes	No tasting or swallowing of any substance in the lab. Goggles on for the demonstration. Teacher alone handles the vinegar-antacid demonstration; use the whole tablet F and dilute vinegar only. Tablets are never taken by learners. Handle the topic of substance abuse sensitively: do not ask learners to share personal or family experiences, and refer anyone who is worried to the school guidance and counselling teacher. Adult doses must never be given to children.

B. LESSON OVERVIEW

Mrs Achieng returns to the Kisumu tray. Wanjiru asked earlier why antacid F relieves heartburn but paracetamol G relieves pain, and why too much of either is dangerous. Learners predict what happens in the body with each tablet, then gather evidence: a teacher-led demo of F fizzing in dilute vinegar (acid) while G does not (a Lesson 2 property), and a data card with the paracetamol label dose. They calculate $8 \times 500 \text{ mg} = 4000 \text{ mg} = 4 \text{ g}$ as the daily maximum, and examine the effects of antacid overuse. They define drug, prescription, dosage and substance abuse, explain why the dose matters, add to the Driving Question Board, and design a poster to sensitise peers and the community. The final step revises the class model with ideas about safe use.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase  VIDEO: Using units to solve problems: Drug dosage <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Crossing the Line: reading informational text: Performance-Enhancing Drugs in Sports 8 <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum)  Search ARES for similar readings	Mrs Achieng holds up antacid tablet F and paracetamol tablet G from the Group 1 tray (Otieno, Wanjiru, Amina and Kipchirchir). Learners recall Wanjiru's puzzle. Individually, in their notebooks, they write predictions for 3 questions: (1) What do you think each tablet does inside the body? (2) Do you think F and G work in the same way? Why or why not? (3) What might happen if a person takes far more tablets than the label says? Then each group of four shares and records one group prediction on a sticky note, with each member's idea acknowledged (Respect).	Say: 'Last week Wanjiru asked why the antacid and the paracetamol are different. Today we find out, and we also find out how to use them safely. Write your own prediction first, in silence, for 2 minutes.' Then: 'Share with your group. Each person speaks once, and the group writes down every idea, even ones you disagree with.' WAIT TIME 10-15 seconds after each question before calling on anyone. Cold-call 3 learners from different groups: 'Tell us one thing your group predicted. Do not tell us whose idea it was, just the idea.' Accept all ideas without judging; write key predictions on the board under 'Our predictions'. Do not reveal the answers yet. Remind: no tasting, tablets stay on the tray.	Predict-Observe-Explain (POE) with individual then group prediction. Writing predictions before evidence exposes prior ideas (for example, that all tablets 'just cure sickness' or that more tablets mean faster relief) so that evidence in the Observe Phase can confirm or challenge them. It links directly to the anchoring phenomenon, where F and G look similar but behave differently.	Look for: (1) every learner has a written prediction before group talk (scan the notebooks during the silent 2 minutes and count empty pages); (2) group sticky notes contain at least 2 ideas; (3) at least 3 of 8 groups mention that a larger amount could be harmful; (4) learners listen without interrupting while others speak. Note common misconceptions on the board: 'all medicines do the same thing', 'more is always better'.
Observe Phase  VIDEO: Using units to solve problems: Drug dosage <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Crossing the	Part A (teacher demo, 5 minutes): Learners watch Mrs Achieng drop a whole antacid tablet F into dilute vinegar (acid) and a whole paracetamol tablet G into dilute vinegar, wearing goggles. F fizzes; G does not fizz. Learners recall from the Lesson 2 water test that F crumbles into a cloudy white suspension while G stays mostly intact and crumbles slowly. Part B (data card, 10 minutes): Each group receives a Data Card with the facts: (i) adult paracetamol label: 1 to 2 tablets of 500 mg every 4 to 6 hours, maximum 8	Before the demo say: 'Goggles on. Only I handle the vinegar and the tablets. Watch the surface of the liquid and tell me what you see, not what you think.' Drop F, then G in separate beakers. WAIT TIME 10-15 seconds, then ask: 'What did you observe in the beaker with F? And with G?' Cold-call 3 learners. Ask: 'Vinegar is an acid. What does the fizzing tell us about how F behaves with acid?' Guide: F reacts with acid, which is why it can neutralise excess stomach acid; G does not react, so it does not neutralise acid. During the data	Evidence table with data interpretation (Analysing and Interpreting Data, NGSS SEP). Placing observations (fizzing or not), label numbers and effects in one table lets learners see patterns: the same property tested in Lesson 2 (reaction with acid) explains the medicine's action, and the dosage numbers explain the safety limits. The calculation turns an abstract warning into a concrete limit of 4 g in 24 hours.	Check that: (1) the table records F fizzes in vinegar and G does not; (2) at least 6 of 8 groups correctly get $8 \times 500 \text{ mg} = 4000 \text{ mg} = 4 \text{ g}$; (3) learners convert mg to g correctly; (4) groups can state the effects of too much of each tablet (liver damage for paracetamol; upset body salts, constipation or diarrhoea for antacid); (5) learners use the words 'maximum' and 'label' in their explanations. Use a thumbs-up check: 'Show me thumbs up if you can explain why 12 tablets in a day is too many.' Tally the number of thumbs.

Line: reading informational text: Performance-Enhancing Drugs in Sports 8 <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum) Search ARES for similar readings	tablets in 24 hours; (ii) paracetamol: more than the label can damage the liver; (iii) antacid: taken too often can upset the body's salts and cause constipation or diarrhoea; (iv) do not give children adult doses; (v) the Pharmacy and Poisons Board regulates medicines in Kenya. Learners complete a table: tablet, what it does, what happens in excess, and calculate the maximum daily mass of paracetamol: $8 \times 500 \text{ mg} = 4000 \text{ mg} = 4 \text{ g}$. They also calculate how many tablets would be taken in 24 hours at 2 tablets every 4 hours (6 doses $\times 2 = 12$ tablets, which is 6000 mg, more than the 4000 mg maximum) to see how easy it is to exceed the limit.	card activity circulate to every group and ask: 'Where did the 4000 mg come from? Show me the working.' and 'Why is 4000 mg the same as 4 g?' ($1000 \text{ mg} = 1 \text{ g}$). Prompt groups that reach 12 tablets: 'Does this follow the label? What does the label's maximum protect?' Remind them that this must never be tried and that only a health worker's advice or the label guides dosage.		
Explain Phase  VIDEO: Using units to solve problems: Drug dosage <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum) Search ARES for similar videos  EXERCISE: Crossing the Line: reading informational text: Performance-Enhancing Drugs in Sports 8 <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum) Search ARES for similar	Learners build a group explanation answering Wanjiru's puzzle in 4 to 5 sentences: (1) why the antacid relieves heartburn but paracetamol relieves pain, using the acid-reaction evidence; (2) why taking too much of either is dangerous, using the dose numbers. Then, using a class discussion, learners build definitions of the four key terms: drug (a substance that changes how the body works), prescription (a health worker's written instruction for a medicine), dosage (amount and timing), and substance abuse (using a substance in a harmful way or amount). They sort 6 short Kenyan scenario cards into 'correct use' or 'misuse' and explain each choice: e.g. (1) an adult follows the label of paracetamol 500 mg, taking 2 tablets every 6 hours; (2) a learner takes 10 paracetamol tablets in a day because a headache is severe; (3) a parent gives a child	Say: 'Use evidence, not opinion. Start with: We saw that F... so it can... G did not... so it...' After groups write, WAIT TIME 10-15 seconds, then cold-call 4 groups to read their explanation and ask the others: 'What evidence in this explanation convinces you? What is missing?' Draw out the key idea: the antacid works because it reacts with and neutralises acid, paracetamol works on pain and fever by a different action, and both are harmful in excess. Give the four definitions only after learners have proposed their own; ask: 'Which word in our definition of substance abuse tells us a medicine can be misused?' (answer: harmful way or amount). For the scenario cards, cold-call 6 different learners, one per card: 'Is this use or misuse? Which evidence from today tells you?' Be sensitive: 'We talk about situations, not about anyone in this room. If	Claim-Evidence-Reasoning (CER) with scenario sorting. Learners make a claim (misuse or correct use), support it with evidence (label dose, 4 g maximum, effects), and give reasoning (properties and actions of the substance). Applying the same reasoning to new cases transfers the idea from the tray to everyday Kenyan life and shows that the properties-and-dose idea is general.	Look for: (1) explanations that include both parts: F reacts with acid and neutralises it; G acts on pain and fever and does not neutralise acid; (2) correct use of the terms drug, prescription, dosage and substance abuse; (3) at least 5 of 6 scenarios correctly classified with a reason that cites the label, dose or effects; (4) learners link to Lesson 8 by pointing to the missing label or expiry date in scenario 5; (5) respectful turn-taking and building on others' ideas. Ask 2 groups to restate a classmate's idea before adding their own.

readings	an adult dose; (4) a person takes antacid tablets all day after every meal for weeks; (5) a person buys medicine from a stall with no label or expiry date (linking to Lesson 8); (6) a person takes medicine prescribed by a clinic exactly as written. Learners also discuss the consumer's right to information on prescriptions, substance use and side effects (Citizenship).	you are worried about yourself or someone you know, see the guidance and counselling teacher.' Briefly stress that the Pharmacy and Poisons Board regulates medicines and that a prescription exists because a health worker chooses the medicine and dose for the person.		
Driving Question Board (DQB) Creation  VIDEO: Using units to solve problems: Drug dosage <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Crossing the Line: reading informational text: Performance-Enhancing Drugs in Sports 8 <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum)  Search ARES for similar readings	Learners return to the Driving Question Board. Each group reads the questions from earlier lessons and decides which can now be answered fully or partly using Lesson 9 evidence (for example Wanjiru's question about why antacid and paracetamol differ, and the question on why too much is dangerous). They move answered questions to the 'Can answer now' column, write the evidence on a sticky note (such as 'F fizzes with acid, G does not' or '8 x 500 mg = 4 g maximum in 24 hours'), and add at least one new question, such as 'How do chemists test that a medicine is safe?' or 'Why do some medicines need a prescription?'.	Say: 'Look at our board. Which questions can we answer better today? Place your sticky note with the evidence next to the question.' WAIT TIME 10-15 seconds before groups move notes. Cold-call 3 groups: 'Which question did you answer, and what is your evidence?' Ask: 'What new question has today raised?' and add 2 to 3 new questions to the board. Link back to the KICD inquiry questions: 'How are medicines for chronic diseases developed? Today we only started: we saw why dosage matters. We will keep this question on the board.' Do not answer every question; mark open ones with a star for Lesson 10.	Driving Question Board as a tracking tool. Matching evidence to questions makes growth in understanding visible and shows learners that science is a continuing process: some questions are answered, others lead to new ones. It keeps the whole class connected to the anchoring phenomenon and the driving question.	Check that: (1) each group moves at least 1 question with a sticky note naming specific evidence from today; (2) evidence is accurate (the numbers 500 mg, 4 g, and F fizzes with acid); (3) each group adds at least 1 relevant new question; (4) learners can say what still needs to be answered. Tally the questions answered versus open on the board.
Model Building Phase  VIDEO: Using units to	Part A (model revision, 8 minutes): Using class model v2 (built in Lesson 6) as the starting point, each group adds a section on safe use: properties of a substance	Say: 'Our model must now answer: how do we use medicines safely? Add what we learned today to model v2. Show how the property of a medicine leads to its use and	Cumulative model revision and Communicating Information (NGSS SEP). Adding safety ideas to the model shows that the properties of matter, the action of a	Observable indicators: (1) the revised model has at least 2 new connections, including property to action and dose to safety; (2) every poster has the paracetamol

solve problems: Drug dosage <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum) Search ARES for similar videos EXERCISE: Crossing the Line: reading informational text: Performance-Enhancing Drugs in Sports 8 <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum) Search ARES for similar readings	decide what it does; the dose decides whether it helps or harms; labels, prescriptions and correct dosage keep people safe. They draw an arrow from the properties of F and G on the tray to 'how the medicine works' to 'safe dose' to 'harm if misused'. Part B (poster project, remaining time and continued as homework): In groups of four, learners plan a poster to sensitise peers and the community on the risks of drug and substance use. Each poster must include: a clear title, 3 facts (one must be the paracetamol maximum of 4 g, i.e. 8 tablets of 500 mg, in 24 hours), an effect of antacid overuse, a message about following the label, prescription and health worker's advice, and where to get help (a health worker or the guidance and counselling teacher). Each member takes a role (ideas, drawing, wording, presenting), and the group agrees on the design by listening to every idea (Respect). Posters are displayed in the school corridor and shared with the community in Kisumu.	how the dose leads to safety.' WAIT TIME 10-15 seconds, then cold-call 2 groups to explain one new arrow in their model. For the poster: 'Your poster must teach someone who was not in this lesson. Use one correct fact from our evidence. No scary pictures of people, and no names of real people.' Circulate and ask: 'How does your poster help a parent in Kisumu avoid giving a child an adult dose?' and 'Who made a decision in your group? Whose idea did you use?' Give the assessment criteria: accuracy, clarity, message, teamwork. Keep drafts for Lesson 10, when posters will be combined with the final model. Remind learners that all the lab rules (no tasting, goggles on) remain in force.	drug and the dosage are linked parts of one explanation. Designing a poster requires learners to select accurate evidence and communicate it to a real audience, which deepens understanding and builds the Communication and Collaboration and Citizenship competencies.	maximum ($8 \times 500 \text{ mg} = 4 \text{ g}$ in 24 hours), an antacid overuse effect and a safety message; (3) all 4 members have a role and have contributed; (4) learners explain their choices using evidence; (5) teacher uses a simple 3-point checklist (accurate, clear, teamwork) per group. Collect a one-sentence exit ticket: 'One thing I will do to use medicines safely is...'
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D. TEACHER REFLECTION

After the lesson, reflect: (1) Did learners connect the property of reacting with acid (Lesson 2) to the antacid's action, rather than simply memorising it? (2) How many groups correctly worked out $8 \times 500 \text{ mg} = 4 \text{ g}$ and used it in their poster? (3) Did any learners still believe 'more medicine means faster relief'? What evidence changed their minds? (4) Was every learner's voice heard in the groups (Respect)? (5) Was the discussion of substance abuse handled sensitively, and does any learner need follow-up with the guidance and counselling teacher? (6) Which new DQB questions should be given priority in Lesson 10? Plan to display posters and check them for accuracy before sharing with the community.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	We observed that antacid F fizzes in dilute vinegar (acid) while paracetamol G does not; both are tablets that behave differently (F crumbles into a cloudy suspension in water, G stays mostly intact). We read from the label that the adult paracetamol maximum is
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	8 tablets of 500 mg in 24 hours, and that antacid taken too often can cause constipation or diarrhoea.
What did I learn?	We learned that a drug is a substance that changes how the body works, a prescription is a health worker's written instruction for a medicine, dosage is the amount and timing, and substance abuse is using a substance in a harmful way or amount. $8 \times 500 \text{ mg} = 4000 \text{ mg} = 4 \text{ g}$ is the paracetamol maximum in 24 hours; more can damage the liver. Children must not be given adult doses.
How does this explain the phenomenon?	The antacid relieves heartburn because it reacts with and neutralises excess stomach acid, while paracetamol relieves pain and fever by a different action and does not neutralise acid. Both are harmful in excess, so we follow the label, prescription and a health worker's advice, and we share this message in posters to keep our school and community safe.

LESSON 10 (80 minutes (double lesson)): Safe and Healthy Learning Environment and the Final Model: How Properties of Substances Shape the Products and Medicines We Use

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	Learners finalise the class model and write a Final Explanation answering the driving question, and brainstorm their rights and responsibilities for a safe and healthy learning environment.
Knowledge	Learners will be able to: (a) explain that substances A-H behave differently because each has its own properties (solubility, magnetism, density, melting and burning behaviour, reaction with acid) which come from what it is made of; (b) state how chemists use such properties to choose, separate and make materials, including medicines; (c) state the rights and responsibilities of learners for a safe and healthy school and laboratory environment; (d) explain how labels, correct dosage and avoiding substance misuse keep people safe.
Skills	Developing and using models (comparing model v1 with the final model); constructing explanations (writing a Final Explanation using claim, evidence and reasoning); engaging in argument from evidence; obtaining, evaluating and communicating information (poster gallery); Communication and Collaboration; Citizenship (critical and constructive dialogue on rights and responsibilities).
Attitudes	Respect: learners show acceptance while appreciating the opinion of each member of the group. Learners advocate for a safe and healthy learning environment and take responsibility for their own and others' safety.
Key Inquiry Question	1. How are medicines for chronic diseases developed? 2. Do all substances around the world have the same properties and uses? 3. Why are food products, detergents and the things in the surrounding especially those that are man-made the way they are?
Purpose in Storyline	Lesson 10 closes the storyline. Learners use the evidence from Lessons 1-9 (the Kisumu tray A-H, the DQB, model v2 and the Lesson 9 posters) to finalise the model and answer the driving question. Learners also link the lab safety rules they have followed all along (goggles on, no tasting, heating by the teacher or supervised) to their rights and responsibilities for a safe and healthy school environment (PCI: life skills, safety and security).
Safety Notes	No new practical heating is done in this lesson. Evidence is revisited from recorded results and labelled samples only. Reinforce lab rules: goggles on whenever chemicals or heat are present, no tasting or smelling directly, no heating without the teacher or a supervisor, wash hands after handling substances. Medicine packs used as props must be empty and clean. Do not let learners handle real tablets for tasting, and remind them that adult doses are not for children. Remind learners that Lesson 9 stated the maximum paracetamol dose is 8 x 500 mg = 4 g in 24 hours and that more can damage the liver.

B. LESSON OVERVIEW

Learners return to the Kisumu school lab tray with eight substances (A table salt, B water, C sand, D sulphur, E iron filings, F antacid tablet, G paracetamol tablet, H sugar). In this final lesson they first predict what a safe and healthy school needs and what the class model will look like, then gather evidence by reviewing the results recorded in Lessons 1-9 (the water test, heat test and the mixture M separation: 2.9 g iron, 2.8 g sand, 2.7 g salt, total 8.4 g recovered out of 9 g, loss 0.6 g). They brainstorm and present their rights and responsibilities for a safe and healthy learning environment and tie these to the safety rules used with the tray. Learners then complete the Driving Question Board by sorting questions into answered and partly answered, and compare model v1 (Lesson 3) with model v2 (Lesson 6) to build the final model. The lesson ends with each learner writing a Final Explanation that answers the driving question: How do the different properties of substances around us help chemists make medicines, foods and other products, and how do we use them safely?

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase  VIDEO: <i>No closely matching video in the ARES library for this activity.</i> Search ARES for videos Search terms: chemistry rights responsibilities safe  HTML: Chemistry and the Environment (Flexbook) <i>Related topic: not an exact match for this lesson.</i> Source: CK-12 Search ARES for similar readings	Learners sit in their Lesson 1 groups of four (Group 1: Otieno, Wanjiru, Amina and Kipchirchir; 8 groups, each with a tray). The teacher shows the tray A-H and the original DQB. In their notebooks, individually for 3 minutes, learners answer two prompts: (1) 'Without looking at your notes, what is the one idea that best explains why A-H behave so differently in water and heat?' (2) 'In our school and laboratory, what do I have a right to, and what must I take responsibility for so that everyone stays safe and healthy?' They then draw a quick sketch or list of what they think the final model should show (substances, properties, uses, safety). They share predictions in their group and note where they disagree.	Hold up the tray and say: 'This is the tray Mrs Achieng set up in Kisumu on day one. Today we finish our story. Before you open any notes, write what you think now.' Give 3 minutes of silent writing. WAIT TIME: after asking 'What one idea explains why A-H behave so differently?', wait 10-15 seconds in silence before taking answers. Cold-call 4 learners from different groups (not volunteers) to read their prediction aloud; record keywords on the board without correcting. Ask: 'Wanjiru asked early on why the antacid relieves heartburn but paracetamol relieves pain. What do you predict is the answer now?' Cold-call 2 more learners. Say: 'Your predictions are your starting point; we will check them against the evidence we have collected.' Circulate and look for learners who mention properties, dosage or labels.	Elicit prior knowledge through individual prediction followed by Think-Pair-Share. Writing before talking makes each learner commit to an idea, so that when the evidence is reviewed they can see how their thinking has changed. Recording predictions on the board gives the class a visible 'before' picture to compare with the Final Explanation.	Look for: (1) at least 3 of 4 learners per group write a prediction that mentions properties or what substances are made of; (2) learners name at least one right (e.g. clean water, safe equipment, information on medicine) and one responsibility (e.g. following lab rules, not misusing substances); (3) learners link the antacid and paracetamol difference to different actions. Note learners who say 'they are just different' without reason, and revisit this in the Explain phase. Tick a class checklist for each group's participation.
Observe Phase  VIDEO: <i>No closely matching video in the ARES library for this activity.</i> Search ARES for videos Search terms: chemistry rights responsibilities safe  EXERCISE: Chemistry and the Environment Practice <i>Related topic: not</i>	Learners gather evidence in two parts. Part A (about 12 minutes, evidence review): each group receives a recorded results card, a copy of their Lesson 2 and 3 data on the WATER TEST, HEAT TEST and mixture M separation, and the Lesson 9 posters on the wall. Using a table with columns for A-H, learners record at least one property for each substance: A salt dissolves completely, clear colourless solution, crackles on heating, no melting; B water; C sand sinks, does not dissolve, no change on heating; D sulphur floats, not wetted, melts at about 115 C then burns with a blue flame	Part A: Say: 'You are scientists reviewing your own data. Find the evidence, do not rely on memory.' Walk around with a checklist and ask each group: 'Which property lets you separate the iron from mixture M? Which property tells you the salt is still in the water?' WAIT TIME: wait 10-15 seconds after each question before prompting. Cold-call 3 groups to report the recovered masses and the loss calculation, $9 - 8.4 = 0.6$ g, and ask 'Why is the total recovered less than 9 g?' Part B: Say: 'Our school is a place where everybody has the right to learn safely. With every right comes a responsibility.	Evidence table (data review) plus Gallery Walk with Stars and Wishes. The table makes learners return to the recorded observations as the source of their claims, not memory. The gallery walk allows learners to compare their ideas on rights and responsibilities with those of other groups and to practise critical and constructive dialogue (Citizenship).	Look for: (1) completed tables with a correct property for all 8 substances and correct names A-H (e.g. calls A 'table salt', H 'sugar'); (2) correct use of 'dissolves' for A and H and 'does not dissolve' for C, D, E; (3) correct arithmetic $2.9 + 2.8 + 2.7 = 8.4$ g and $9 - 8.4 = 0.6$ g; (4) each group lists at least 3 rights and 3 responsibilities, and links at least one to a lab rule; (5) stars and wishes are specific and respectful (observable respect indicator: members wait for each other and acknowledge ideas). Record on a rubric: meets, approaching, not yet.

<i>an exact match for this lesson.</i> Source: CK-12 Search ARES for similar readings	and a sharp choking smell; E iron filings sink, pulled out by a magnet, glow orange-red and turn dull blackish; F antacid crumbles into a cloudy white suspension and fizzes in dilute vinegar or lemon juice; G paracetamol stays mostly intact, crumbles slowly, no fizz, melts at about 169 C and browns; H sugar dissolves, melts at about 186 C, browns and chars with a caramel smell. They re-check mixture M: 3 g iron filings + 3 g sand + 3 g salt (9 g); magnet removes iron (2.9 g), filtration removes sand (2.8 g), evaporation recovers salt (2.7 g); total 8.4 g, loss 0.6 g from spillage and sticking to equipment. Part B (about 15 minutes, safe and healthy environment): groups brainstorm on sugar-paper their rights and responsibilities for a safe and healthy learning environment, using the lab safety rules practised with the tray (goggles on, no tasting, heating by the teacher or supervised) and the Lesson 8 and 9 ideas on labels, the maximum paracetamol dose of $8 \times 500 \text{ mg} = 4 \text{ g}$ in 24 hours and the risks of misusing substances. Each group then does a 2-minute gallery presentation, and listeners give one 'star' (strength) and one 'wish' (suggestion) using respectful language.	Write both.' Prompt with 'Which lab rule protected you when sulphur burned?' and 'Which responsibility protects a younger child from an adult paracetamol dose?' Remind learners of Respect: 'Listen fully before you respond; credit the idea of another member.' During gallery presentations cold-call 4 listeners to give a star and a wish. Clarify any misconception, for example that a medicine is safe because it is sold in a shop; correct with label, expiry date and dose information. Reinforce: no tasting and goggles on at all times.		
Explain Phase  VIDEO: <i>No closely matching video in the ARES library for this activity.</i> Search ARES for videos Search terms: chemistry rights responsibilities	Learners write and discuss a Final Explanation answering the driving question. Step 1 (5 minutes): in groups, learners build a Claim-Evidence-Reasoning (CER) chart. Claim: substances behave differently because each has its own set of properties that come from what it is made of. Evidence: at least four pieces from the tray	Say: 'A good scientist does not only say what happened, but why. Use Claim, Evidence, Reasoning.' Model one sentence on the board: 'I claim that sand and salt behave differently because ...' and ask learners to finish it. WAIT TIME: after 'Why does the antacid fizz in vinegar but paracetamol does not?' wait 10-15 seconds, then	Claim-Evidence-Reasoning (CER) with a Final Explanation. CER requires learners to connect a general claim about properties to specific evidence from the tray and to the reasoning that links the two, so that the explanation answers the driving question rather than just listing facts. Applying it to the antacid and paracetamol puzzle	Look for: (1) the claim states that properties come from what a substance is made of; (2) at least 3 correct pieces of tray evidence with correct units (g, mL, C, mg); (3) the reasoning links properties to a use (separation of mixture M, choosing medicines, making food and household products); (4) the antacid is explained as neutralising

safe  EXERCISE: Physical and chemical changes <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum) Search ARES for similar readings	(e.g. A and H dissolve and cannot be filtered out; C and E do not dissolve and can be filtered or pulled out by a magnet; D burns with a blue flame while C does not; F fizzes in acid while G does not). Reasoning: chemists use properties to choose, separate and make materials, so mixture M was separated by magnetic separation, filtration and evaporation. Step 2 (5 minutes): learners explain the medicine puzzle, Wanjiru's question: the antacid (F, calcium carbonate with magnesium hydroxide) reacts with and neutralises excess stomach acid, relieving heartburn; paracetamol (G, 500 mg) relieves pain and fever by a different action and does not neutralise acid; both are harmful in excess (paracetamol over 4 g in 24 hours can damage the liver; too much antacid can cause constipation or diarrhoea). Step 3 (10 minutes): each learner individually writes a Final Explanation of 8-10 sentences answering how properties help chemists make medicines, foods and other products, and how we use them safely, and links it to the three KICD key inquiry questions (medicines for chronic diseases, different properties and uses of substances, why food products and detergents are the way they are). Examples to include: Kericho tea, Magadi soda ash, Mumias or Nzoia sugar, Kenyan pharmaceutical manufacturers in Nairobi, KEBS Diamond Mark, Pharmacy and Poisons Board, and ugali or nyama choma as everyday food chemistry.	cold-call 3 learners. Ask: 'Which property did chemists use to separate iron from sand and salt? Which property did we use for salt?' Cold-call 2 more learners. Probe: 'If a friend says, I took six paracetamol tablets at once because my head hurts a lot, what would you tell them?' Cold-call 2 learners to respond using the label (1 to 2 tablets of 500 mg every 4 to 6 hours, maximum 8 tablets in 24 hours) and say that children must not be given adult doses. Press for evidence: 'What data show that? Point to your table.' Circulate during individual writing, and use sentence starters ('because', 'which shows that', 'therefore') for learners who need support. Do not re-teach earlier lessons; refer learners to the branches (Lesson 5), industries (Lesson 6), labels (Lesson 8) and drug and dosage ideas (Lesson 9) as they have been taught.	transfers understanding to a new, real-life situation.	acid and paracetamol as acting on pain and fever, not acid; (5) the explanation includes safe use: reading labels, correct dosage (maximum 4 g of paracetamol in 24 hours), no adult doses for children, avoiding substance misuse. Score with a 4-point rubric (claim, evidence, reasoning, safety). Collect Final Explanations to identify learners who need follow-up.
Driving Question	Learners complete the Driving Question Board. The class DQB	Say: 'Our board is our record of learning. Let us see how far we	Sorting questions on the Driving Question Board (answered, partly	Look for: (1) each sorted question has a correct lesson number and a

Board (DQB) Creation  VIDEO: No closely matching video in the ARES library for this activity.  Search ARES for videos Search terms: chemistry rights responsibilities safe  HTML: Chemistry and the Environment (Flexbook) Related topic: not an exact match for this lesson. Source: CK-12  Search ARES for similar readings	from Lesson 1 (with questions added across lessons) is displayed. Step 1: each group receives 4-5 sticky notes with questions from the DQB and sorts them into three columns: 'Answered', 'Partly answered' and 'New question'. For each answered question, learners write the lesson number and the evidence that answered it (e.g. 'Why can some mixtures be separated easily while others cannot?' answered by mixture M: magnet, filtration, evaporation; 'Why does the antacid relieve heartburn while paracetamol relieves pain?' answered by acid-neutralising action versus action on pain and fever). Step 2: learners post one new question that they still wonder about (e.g. how Kenyan pharmaceutical manufacturers in Nairobi test medicines before sale, or how the Pharmacy and Poisons Board checks them). Step 3: the class agrees on a final sentence that answers the driving question and writes it at the top of the DQB.	have come.' Give each group their notes and say: 'Decide where each question belongs and prove it with evidence from a lesson.' WAIT TIME: after asking 'Which questions are still partly answered?' wait 10-15 seconds. Cold-call 4 groups (one question each) to justify their sorting and ask 'What evidence supports that?' Ask: 'Which of the three KICD key inquiry questions did we answer best? Which one needs more study?' Cold-call 2 learners. Probe respectfully when learners disagree: 'Amina and Kipchirchir see this differently. Each of you, say the other's idea in your own words first.' Record the agreed class answer. Point out that the unanswered questions will be taken up in later strands, such as 1.2 The Atom (How are electrons arranged in an atom?), without teaching that content now.	answered, new) with evidence justification. This turns the DQB from a list of curiosity into a record of learning, and makes learners trace each answer back to specific evidence and lessons, showing the progress of understanding over time.	piece of evidence; (2) learners can distinguish a fully answered question from a partly answered one, with a reason; (3) at least one new, testable question is posted by each group; (4) learners paraphrase peers' ideas before disagreeing (observable Respect). Tally on a checklist how many groups placed 'Why do powders behave differently?' and 'How does the antacid work?' as answered with evidence.
Model Building Phase  VIDEO: No closely matching video in the ARES library for this activity.  Search ARES for videos Search terms: chemistry rights responsibilities safe  EXERCISE: Chemistry and the Environment	Learners finalise the class model. Step 1 (5 minutes): each group displays model v1 (Lesson 3: classifying substances as elements, compounds or mixtures and the first ideas on properties) and model v2 (Lesson 6: revised with the uses in Kenyan industries). Learners list what has changed. Step 2 (10 minutes): groups redraw a final model on a large sheet with 4 parts: (a) what the substances are made of, classified as elements (D sulphur, E iron), compounds (A salt, B water, H sugar, calcium carbonate in F) and mixtures (mixture M;	Say: 'A model is a tool for explaining. Compare v1 and v2 with what you now know, and improve it.' Provide sentence frames for the comparison: 'In v1 we showed ..., in the final model we added ... because the evidence shows ...'. Ask: 'What was missing from model v1 that we needed to explain why the antacid and paracetamol differ?' WAIT TIME: wait 10-15 seconds after this question. Cold-call 4 spokespersons to compare v1 with the final model. Ask: 'Where on your model do safety and labels fit? Why must safety be part of a	Model revision and comparison (v1, v2 to final). Placing the models side by side lets learners see how their understanding grew from sorting and classifying to explaining properties, uses and safety. The sentence frame 'At first we thought..., now we know... because...' forces explicit reasoning about change in ideas, linking the evidence collected to the final explanation of the phenomenon.	Look for: (1) the final model includes all four parts (what substances are made of, properties, uses, safety) with correct classification of A-H; (2) at least two changes from v1 are named and justified with evidence; (3) the model includes the antacid neutralising acid versus paracetamol acting on pain and fever; (4) safety (labels, dosage, rights and responsibilities) is represented, not only chemistry content; (5) spokespersons use scientific vocabulary correctly. Use a 3-level rubric (complete, partial, incomplete) and record who needs

Practice <i>Related topic: not an exact match for this lesson.</i> Source: CK-12 Search ARES for similar readings	tablets are mixtures of an active ingredient with binders); (b) their properties (solubility, magnetism, density, melting and burning behaviour, reaction with acid); (c) how chemists use the properties (separate by magnet, filtration and evaporation; choose and make medicines, foods such as sugar from Mumias or Nzoia, tea from Kericho, soda ash from Magadi, and other products); (d) how we stay safe (labels with the KEBS Diamond Mark, expiry date, ingredients, caution messages; correct dosage, e.g. paracetamol maximum 8 x 500 mg = 4 g in 24 hours; avoiding substance abuse; lab safety rules and learner rights and responsibilities). Step 3 (5 minutes): a spokesperson from each group explains how the final model differs from v1 in a model comparison, completing the sentence 'At first we thought ..., now we know ... because ...'. The class agrees on one final model and displays it beside the DQB.	chemistry model?' Cold-call 2 learners. Check that substances are called by letters A-H and correct names, that only 'filtration', 'magnetic separation' and 'evaporation' are used, and that classification matches the conventions. After the class agrees on the final model, ask each learner to complete the summary table individually and make a personal commitment to one safety responsibility. Close with: 'Chemistry helps us to understand how drugs interact with the body and to recognise the risks of drug abuse. You now have the tools to use that knowledge safely, at school, at home and in your community.'		support. Final check: learners' summary tables are consistent with their Final Explanation.
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D. TEACHER REFLECTION

After the lesson, reflect on: (1) Did learners use properties, not just description, to explain why A-H behave differently? Which groups still say 'they are just different'? (2) Could learners explain why the antacid (F) and paracetamol (G) work differently, and state the maximum paracetamol dose of 4 g in 24 hours with the reason? (3) Did the brainstorm on rights and responsibilities lead to real commitments (for example following lab rules, reading labels, not sharing medicines, not misusing substances), and did learners show Respect when others spoke? (4) Which learners struggled with the claim-evidence-reasoning structure and what support (sentence starters, worked example) helped? (5) How did the final model differ from v1, and did every group include safety? (6) Which DQB questions remained 'partly answered' and how will they be followed up in Sub-Strand 1.2 The Atom? (7) Was wait time of 10-15 seconds used consistently, and did cold-calling give quieter learners, including girls, a voice? Note changes to make next time.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	What we observed: Using the recorded results for A-H, write what you observed. For example, A salt and H sugar dissolve and cannot be filtered out; C sand and E iron filings do not dissolve, and E is pulled out by a magnet; D sulphur melts at about 115 C then burns with a blue flame and a sharp choking smell; F antacid fizzes in vinegar but G paracetamol does not; mixture M gave
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	2.9 g iron, 2.8 g sand and 2.7 g salt (8.4 g recovered, 0.6 g lost).
What did I learn?	What we learned: Write the ideas from Lessons 1-10 that help answer the driving question: each substance has its own properties which come from what it is made of; chemistry studies matter, its composition, properties, changes and uses; chemists use properties to choose, separate and make materials; labels (quality mark, expiry date, ingredients, caution messages), correct dosage and avoiding substance abuse keep people safe; learners have rights and responsibilities for a safe and healthy learning environment.
How does this explain the phenomenon?	What we can now explain: In 8-10 sentences, explain how the different properties of substances help chemists make medicines, foods and other products, and how we use them safely. Include why the antacid relieves heartburn by neutralising acid while paracetamol relieves pain and fever by a different action, why both are harmful in excess (paracetamol maximum $8 \times 500 \text{ mg} = 4 \text{ g}$ in 24 hours), and one responsibility you will take for a safe and healthy environment at school and at home.

DIFFERENTIATION AND INCLUSION		
Learner Need	Support Strategies	Extension Strategies
English Language Learners	Provide bilingual glossaries; allow use of first language for initial model drawing.	Write extended explanations of observations; lead group discussions in English.
Students with learning difficulties	Provide structured note frames; pair with a supportive partner during investigations.	Mentor peers; create additional visual models to explain concepts.
Gifted and advanced learners	Access to additional reading materials; challenge questions during DQB.	Lead model-building discussions; research and present extension topics to class.